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(54) **METHOD AND APPARATUS FOR IMAGE ENCODING/DECODING ON BASIS OF POC MSB INFORMATION, AND RECORDING MEDIUM FOR STORING BITSTREAM**

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(57) **ABSTRACT**

An image encoding/decoding method and apparatus is provided. The image decoding method comprises determining whether first information specifying a value of a picture order count (POC) most significant bit (MSB) cycle of a current picture is present in a picture header, determining a POC MSB of the current picture based on whether the first information is present in the picture header, and deriving a POC of the current picture based on the determined POC MSB. Based on the first information being not present in the picture header and the current picture being not a CLVSS (coded layer video sequence start) picture, a POC difference between the current picture and a previous picture preceding the current picture in decoding order may be less than half a maximum value of a POC least significant bit (LSB) of the current picture.

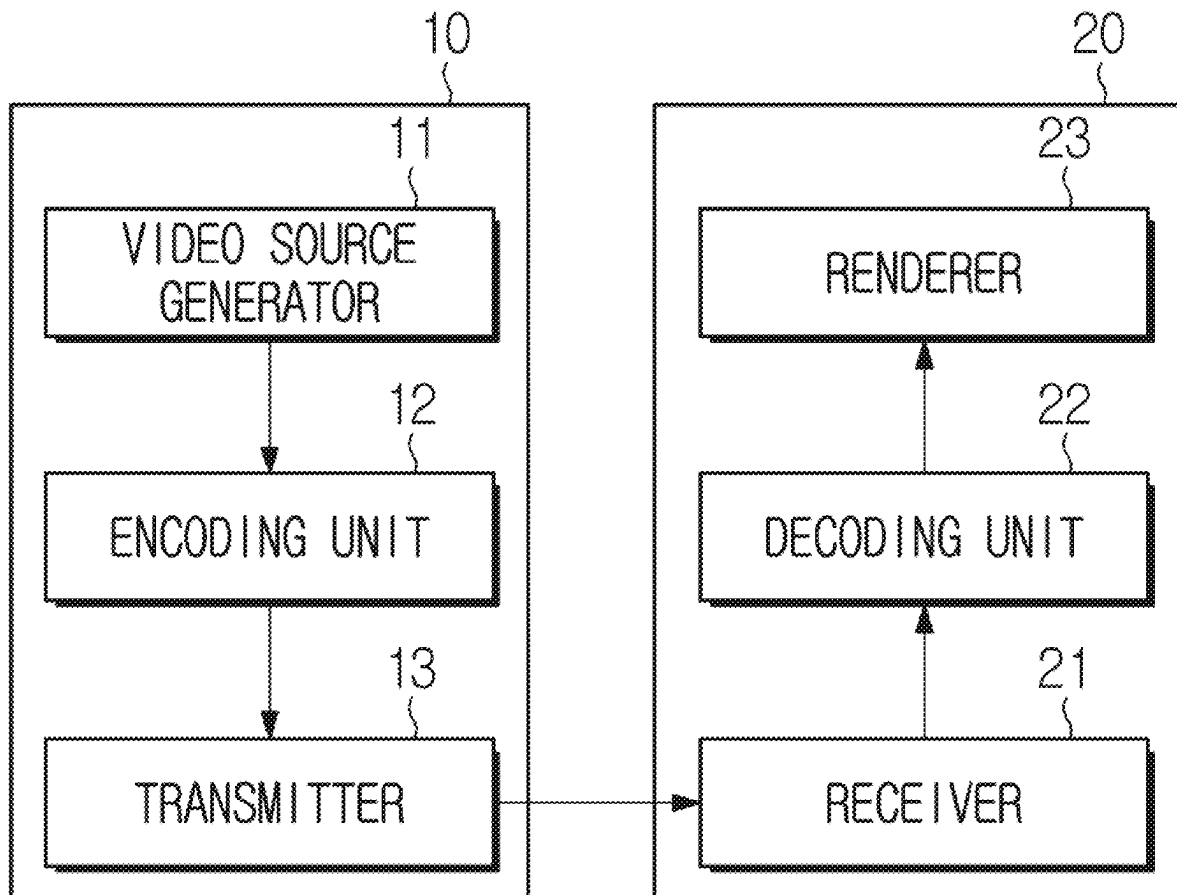


FIG. 1

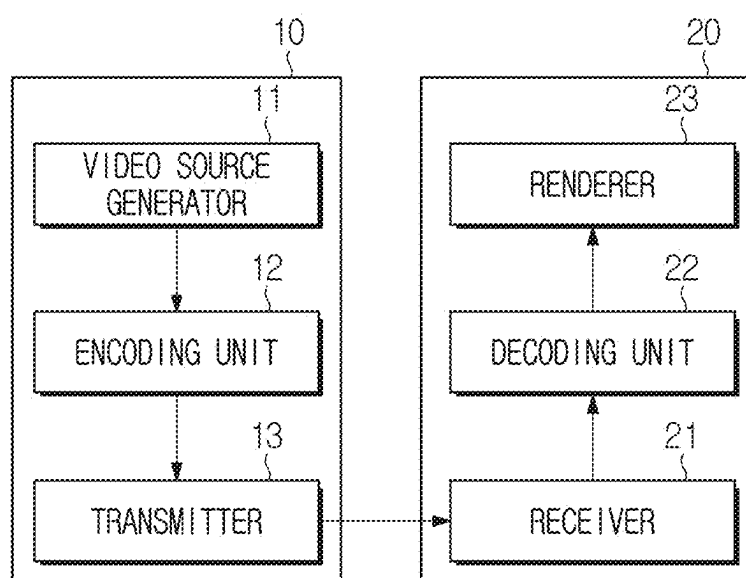


FIG. 2

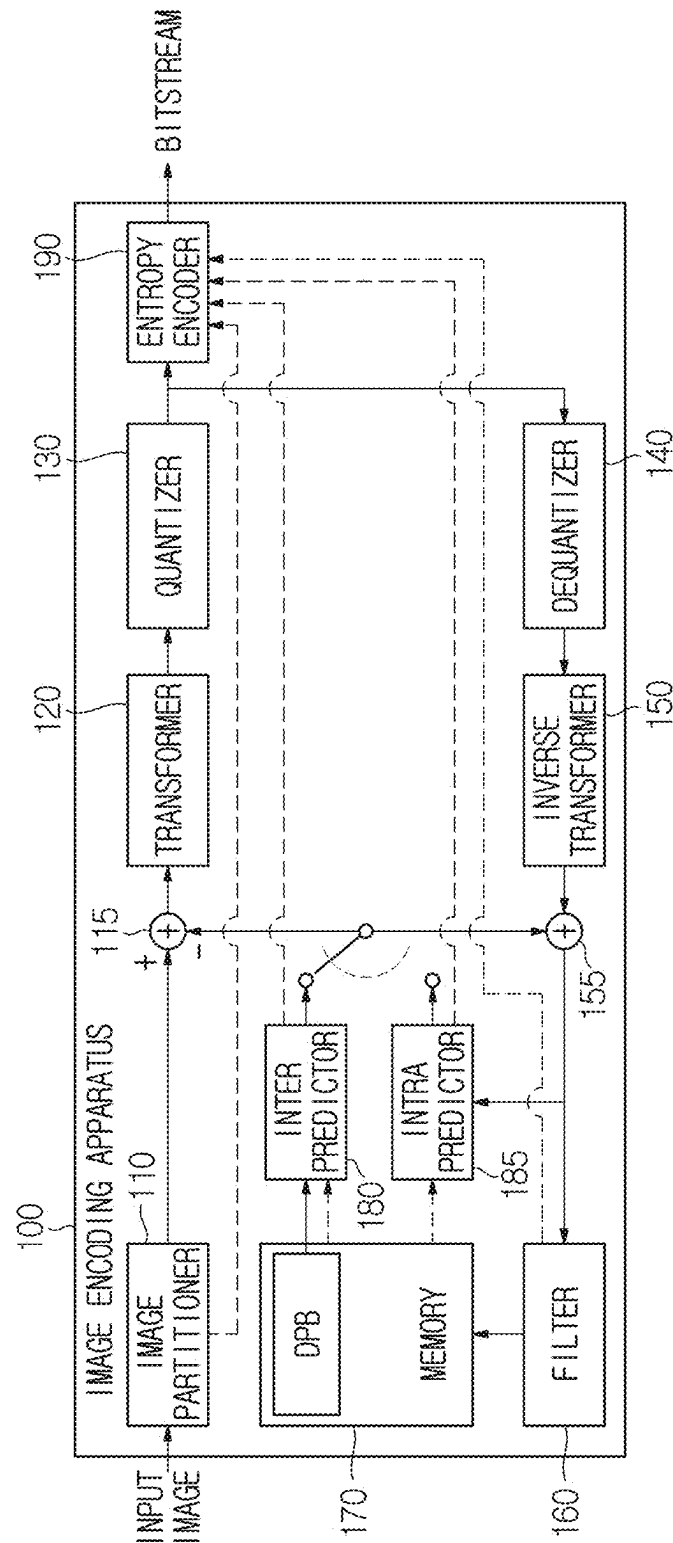


FIG. 3

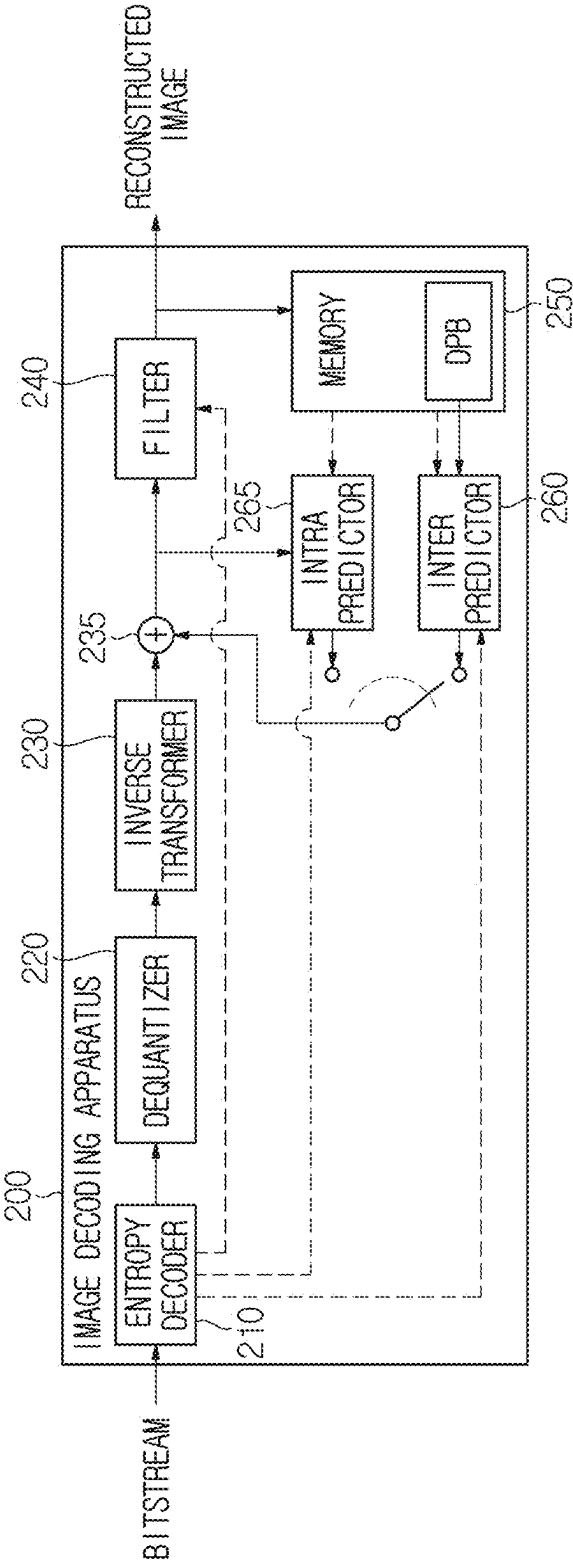


FIG. 4

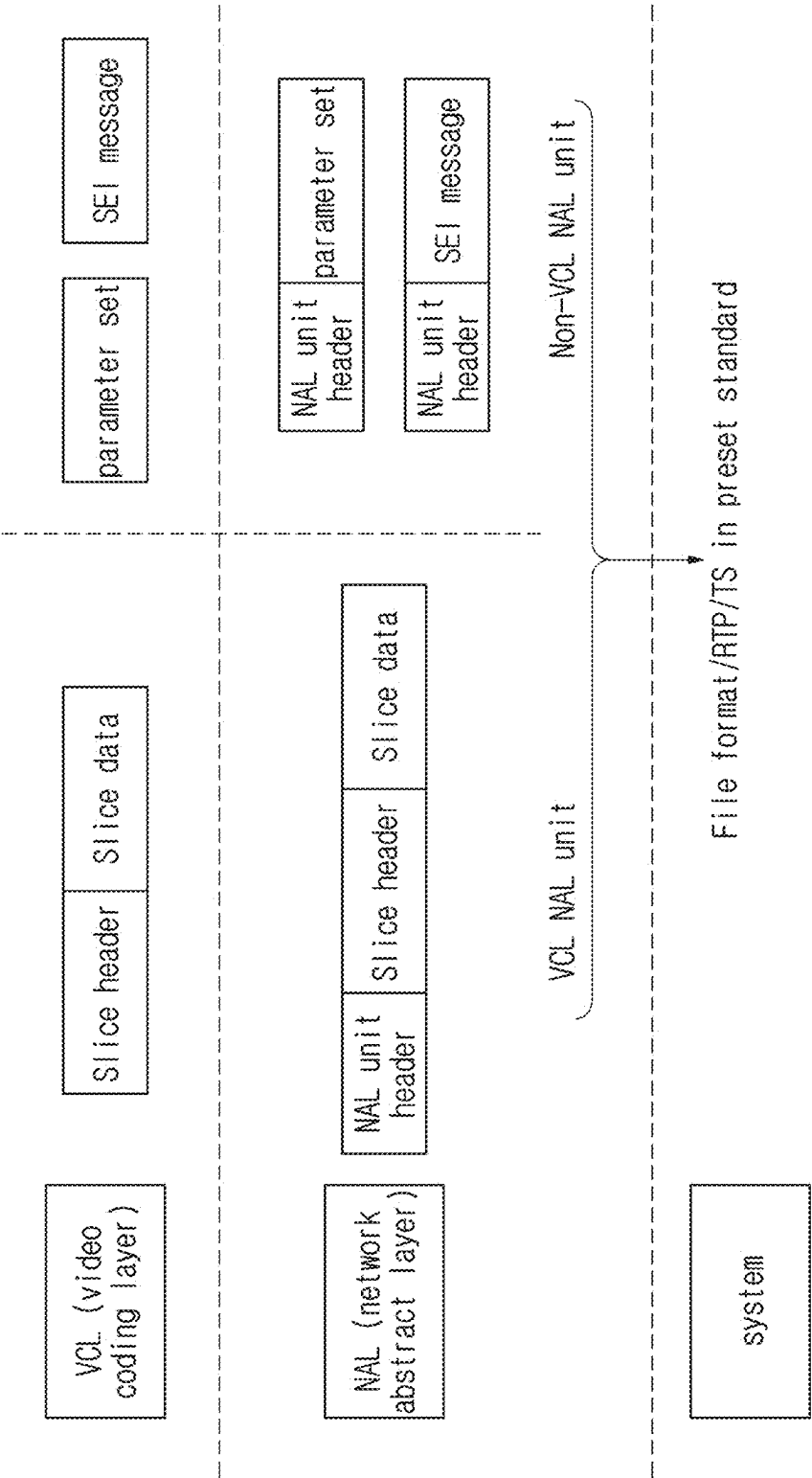


FIG. 5

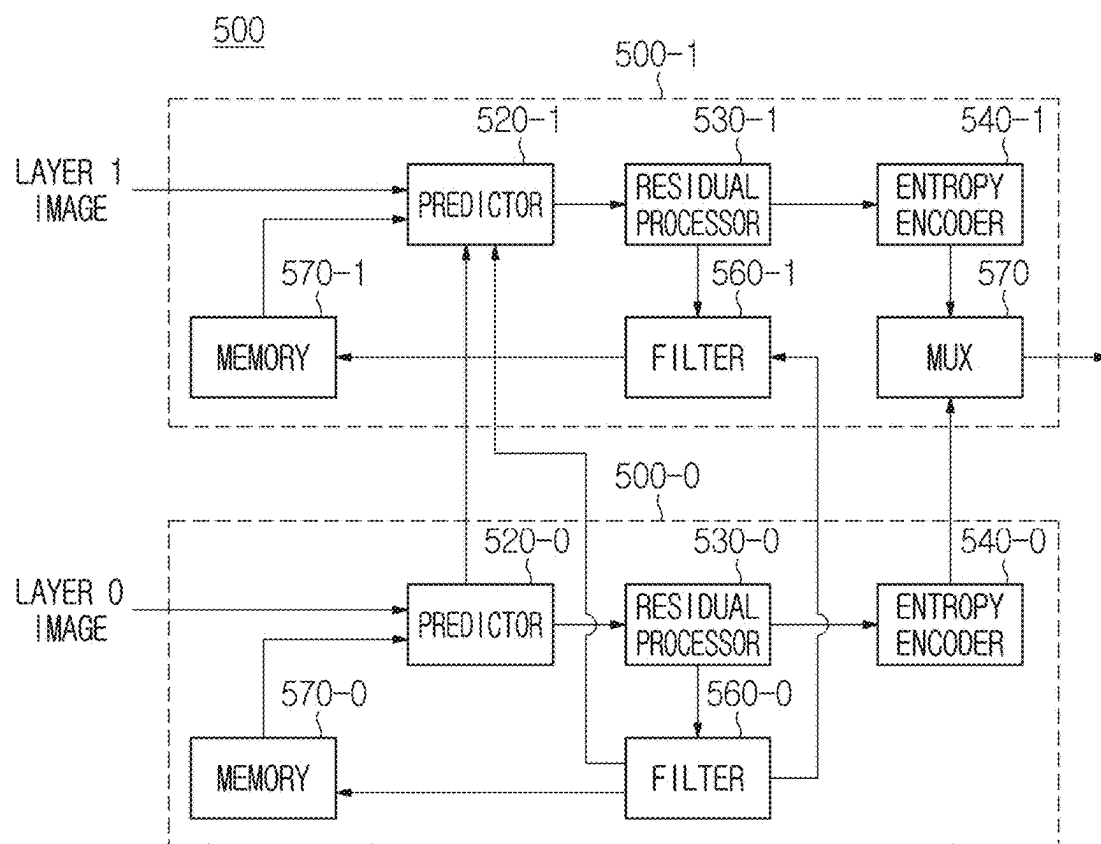


FIG. 6

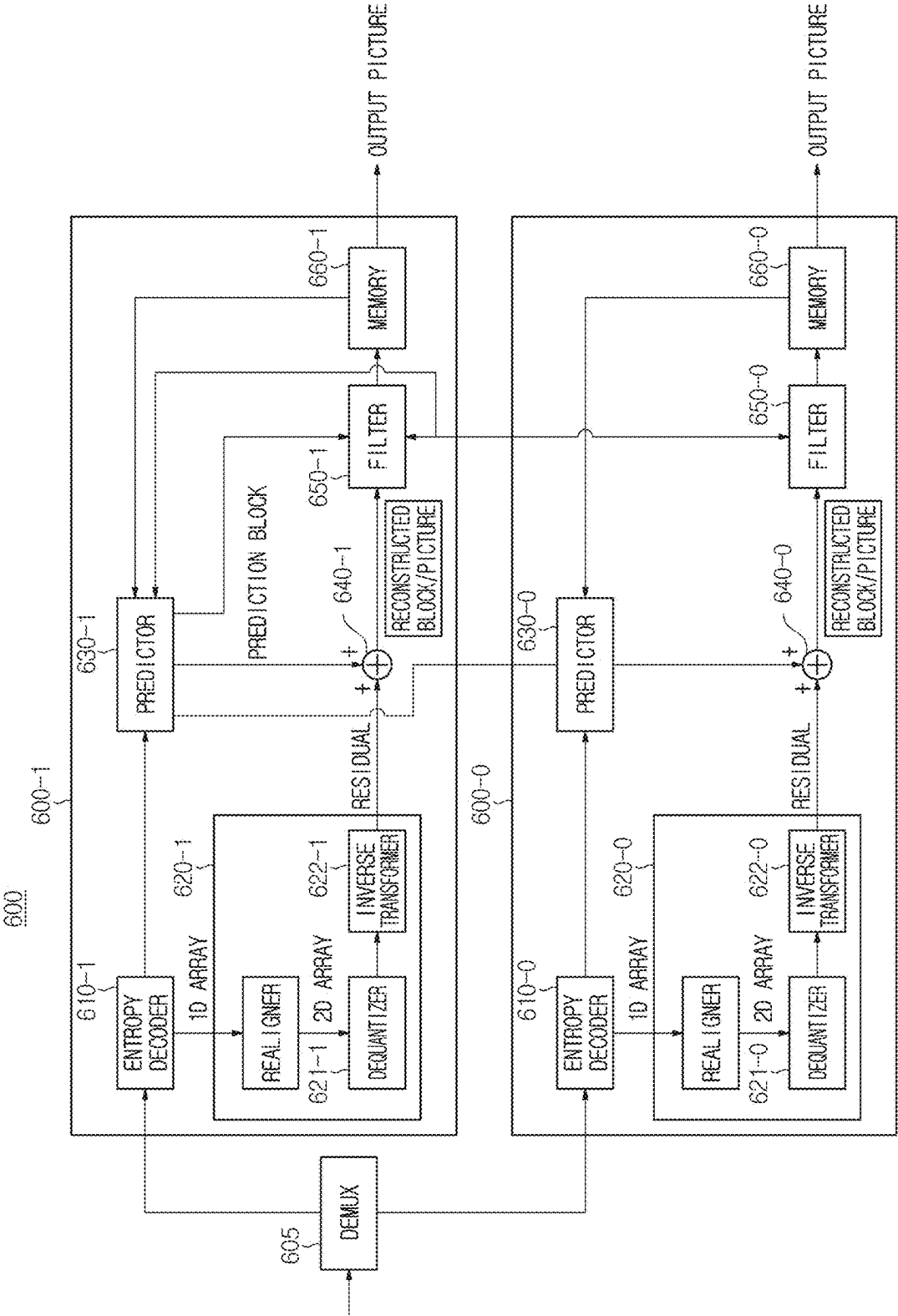


FIG. 7

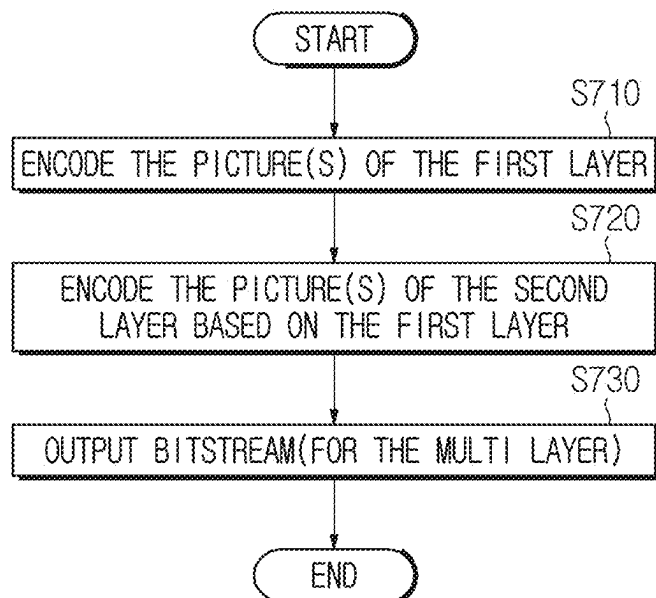


FIG. 8

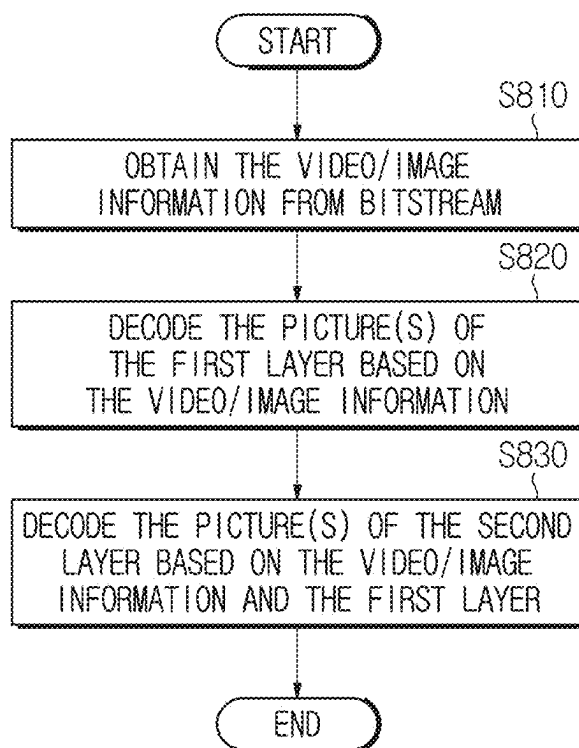


FIG. 9

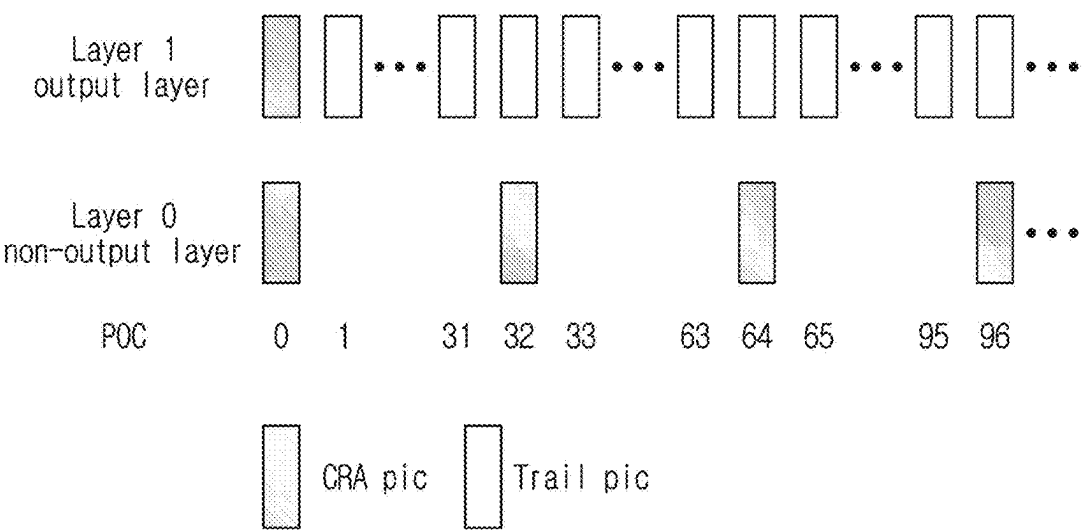


FIG. 10

seq_parameter_set_rbsp() {	Descriptor
sps_seq_parameter_set_id	u(4)
sps_video_parameter_set_id	u(4)
...	
sps_poc_msb_cycle_flag	u(1)
...	
}	

FIG. 11

picture_header_structure() {	Descriptor
...	
if(ph_gdr_pic_flag)	
ph_recovery_poc_cnt	ue(v)
...	
if(sps_poc_msb_cycle_flag) {	
ph_poc_msb_cycle_present_flag	u(1)
if(pli_poc_msb_cycle_present_flag)	
ph_poc_msb_cycle_val	u(v)
...	
}	
}	

FIG. 12

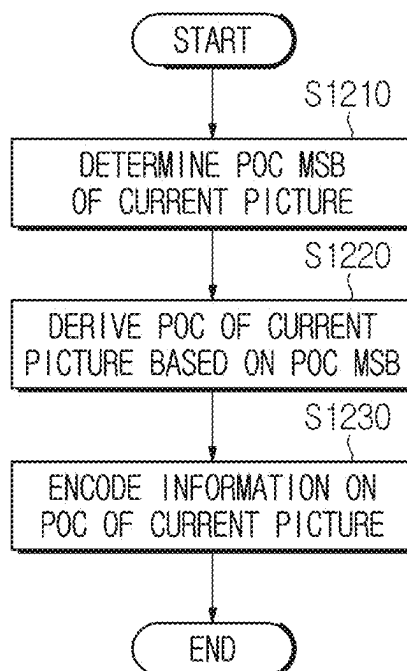


FIG. 13

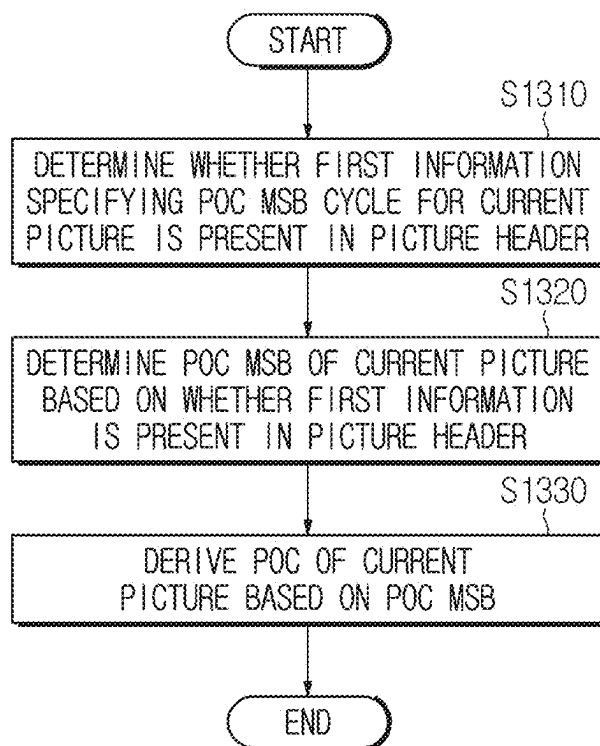
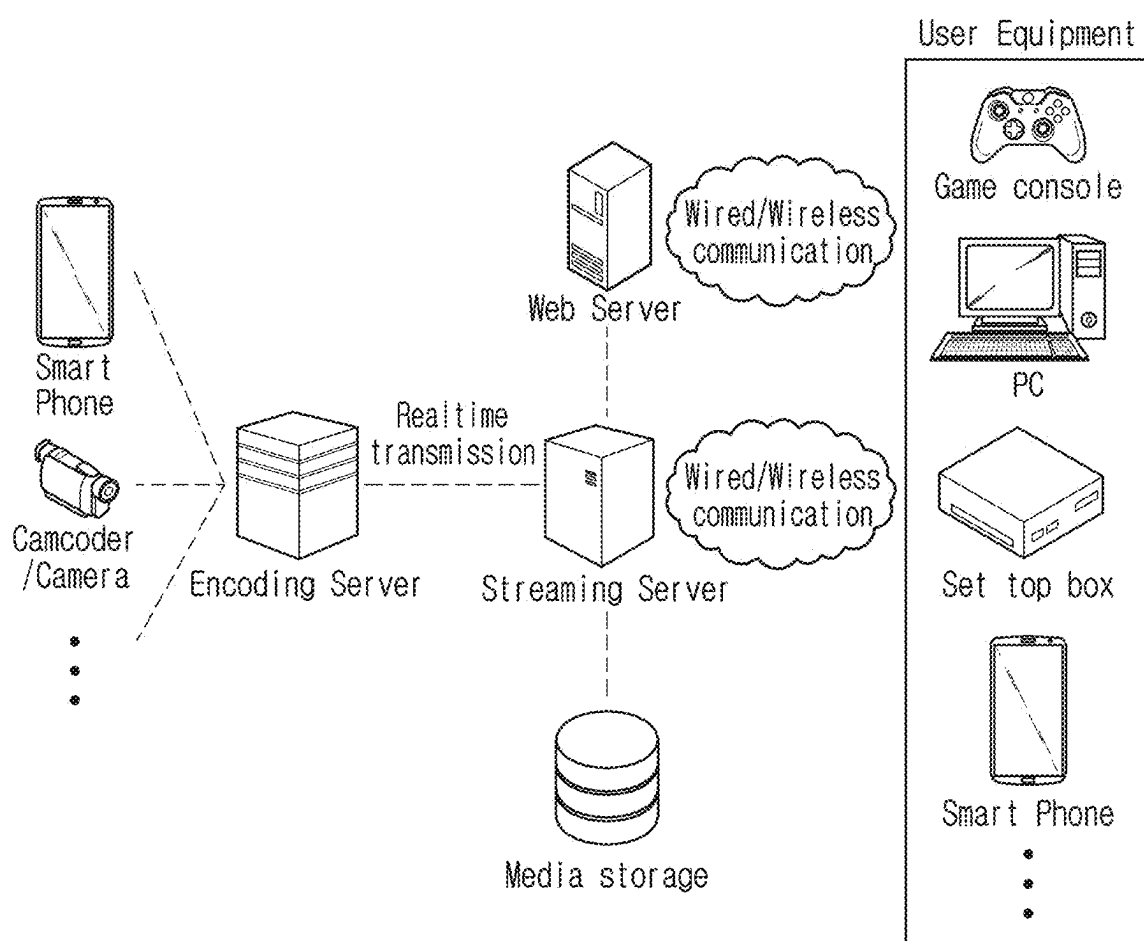


FIG. 14



**METHOD AND APPARATUS FOR IMAGE
ENCODING/DECODING ON BASIS OF POC
MSB INFORMATION, AND RECORDING
MEDIUM FOR STORING BITSTREAM**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This application is a continuation of U.S. application Ser. No. 18/008,619, filed on Dec. 6, 2022, which is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/KR2021/007352, filed on Jun. 11, 2021, which claims the benefit of U.S. Provisional Application No. 63/038,903, filed on Jun. 15, 2020, and U.S. Provisional Application No. 63/039,421, filed on Jun. 15, 2020. The disclosures of the prior applications are incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to an image encoding/decoding method and apparatus and a recording medium storing a bitstream, and, more particularly, to an image encoding/decoding method and apparatus based on POC MSB information, and a recording medium for storing bitstream generated by the image encoding method/apparatus of the present disclosure.

BACKGROUND

[0003] Recently, demand for high-resolution and high-quality images such as high definition (HD) images and ultra high definition (UHD) images is increasing in various fields. As resolution and quality of image data are improved, the amount of transmitted information or bits relatively increases as compared to existing image data. An increase in the amount of transmitted information or bits causes an increase in transmission cost and storage cost.

[0004] Accordingly, there is a need for high-efficient image compression technology for effectively transmitting, storing and reproducing information on high-resolution and high-quality images.

SUMMARY

[0005] An object of the present disclosure is to provide an image encoding/decoding method and apparatus with improved encoding/decoding efficiency.

[0006] Another object of the present disclosure is to provide an image encoding/decoding method and apparatus based on POC MSB information.

[0007] Another object of the present disclosure is to provide an image encoding/decoding method and apparatus based on POC MSB cycle information for which signaling is forced for an intra picture.

[0008] Another object of the present disclosure is to provide an image encoding/decoding method and apparatus based on a POC difference between pictures which is forced to be less than half a maximum value of a POC LSB.

[0009] Another object of the present disclosure is to provide a non-transitory recording medium storing a bitstream generated by an image encoding method or apparatus according to the present disclosure.

[0010] Another object of the present disclosure is to provide a non-transitory recording medium storing a bit-

stream received, decoded and used to reconstruct an image by an image decoding apparatus according to the present disclosure.

[0011] Another object of the present disclosure is to provide a method of transmitting a bitstream generated by an image encoding method or apparatus according to the present disclosure.

[0012] The technical problems solved by the present disclosure are not limited to the above technical problems and other technical problems which are not described herein will become apparent to those skilled in the art from the following description.

[0013] An image decoding method according to an aspect of the present disclosure comprises determining whether first information specifying a value of a picture order count (POC) most significant bit (MSB) cycle of a current picture is present in a picture header, determining a POC MSB of the current picture based on whether the first information is present in the picture header, and deriving a POC of the current picture based on the determined POC MSB. Based on the first information being not present in the picture header and the current picture being not a CLVSS (coded layer video sequence start) picture, a POC difference between the current picture and a previous picture preceding the current picture in decoding order may be less than half a maximum value of a POC least significant bit (LSB) of the current picture.

[0014] An image decoding apparatus according to another aspect of the present disclosure comprises a memory and at least one processor. The at least one processor may determine whether first information specifying a value of a picture order count (POC) most significant bit (MSB) cycle of a current picture is present in a picture header, determine a POC MSB of the current picture based on whether the first information is present in the picture header, and derive a POC of the current picture based on the determined POC MSB. Based on the first information being not present in the picture header and the current picture being not a CLVSS (coded layer video sequence start) picture, a POC difference between the current picture and a previous picture preceding the current picture in decoding order may be less than half a maximum value of a POC least significant bit (LSB) of the current picture.

[0015] An image encoding method according to another aspect of the present disclosure comprises determining a picture order count (POC) most significant bit (MSB) of a current picture, deriving a POC of the current picture based on the determined POC MSB, and encoding information on the derived POC. Based on first information specifying a value of a POC MSB cycle of the current picture being not encoded and the current picture being not a CLVSS (coded layer video sequence start) picture, a POC difference between the current picture and a previous picture preceding the current picture in decoding order may be less than half a maximum value of a POC least significant bit (LSB) of the current picture.

[0016] In addition, a computer-readable recording medium according to another aspect of the present disclosure may store the bitstream generated by the image encoding apparatus or the image encoding method of the present disclosure.

[0017] In a transmission method according to another aspect of the present disclosure, a bitstream generated by an

image encoding method or an image encoding apparatus of the present disclosure may be transmitted.

[0018] The features briefly summarized above with respect to the present disclosure are merely exemplary aspects of the detailed description below of the present disclosure, and do not limit the scope of the present disclosure.

[0019] According to the present disclosure, it is possible to provide an image encoding/decoding method and apparatus with improved encoding/decoding efficiency.

[0020] According to the present disclosure, it is possible to provide an image encoding/decoding method and apparatus based on POC MSB information.

[0021] According to the present disclosure, it is possible to provide an image encoding/decoding method and apparatus based on POC MSB cycle information for which signaling is forced for an intra picture.

[0022] According to the present disclosure, it is possible to provide an image encoding/decoding method and apparatus based on a POC difference between pictures which is forced to be less than half a maximum value of a POC LSB.

[0023] Also, according to the present disclosure, it is possible to provide a computer-readable recording medium storing a bitstream generated by an image encoding method or apparatus according to the present disclosure.

[0024] Also, according to the present disclosure, it is possible to provide a computer-readable recording medium storing a bitstream received, decoded and used to reconstruct an image by an image decoding apparatus according to the present disclosure.

[0025] Also, according to the present disclosure, it is possible to provide a method of transmitting a bitstream generated by an image encoding method or apparatus according to the present disclosure.

[0026] It will be appreciated by persons skilled in the art that the effects that can be achieved through the present disclosure are not limited to what has been particularly described hereinabove and other advantages of the present disclosure will be more clearly understood from the detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a view schematically illustrating a video coding system, to which an embodiment of the present disclosure is applicable.

[0028] FIG. 2 is a view schematically illustrating an image encoding apparatus, to which an embodiment of the present disclosure is applicable.

[0029] FIG. 3 is a view schematically illustrating an image decoding apparatus, to which an embodiment of the present disclosure is applicable.

[0030] FIG. 4 is a view illustrating an example of a layer structure for a coded image/video.

[0031] FIG. 5 is a schematic block diagram of a multi-layer encoding apparatus, to which embodiment(s) of the present disclosure is applicable, and, in which encoding of a multi-layer video/image signal is performed.

[0032] FIG. 6 is a schematic block diagram of a decoding apparatus, to which embodiment(s) of the present disclosure is applicable, and, in which decoding of a multi-layer video/image signal is performed.

[0033] FIG. 7 is a view illustrating a method of encoding an image based on a multi-layer structure by an image encoding apparatus according to an embodiment.

[0034] FIG. 8 is a view illustrating a method of decoding an image based on a multi-layer structure by an image decoding apparatus according to an embodiment.

[0035] FIG. 9 is a view illustrating problems which may occur during POC derivation under a multi-layer structure.

[0036] FIG. 10 is a view illustrating an example of SPS syntax.

[0037] FIG. 11 is a view illustrating an example of picture header syntax.

[0038] FIG. 12 is a flowchart illustrating an image encoding method according to an embodiment of the present disclosure.

[0039] FIG. 13 is a flowchart illustrating an image decoding method according to an embodiment of the present disclosure.

[0040] FIG. 14 is a view illustrating a content streaming system, to which an embodiment of the present disclosure is applicable.

DETAILED DESCRIPTION

[0041] Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the accompanying drawings so as to be easily implemented by those skilled in the art. However, the present disclosure may be implemented in various different forms, and is not limited to the embodiments described herein.

[0042] In describing the present disclosure, if it is determined that the detailed description of a related known function or construction renders the scope of the present disclosure unnecessarily ambiguous, the detailed description thereof will be omitted. In the drawings, parts not related to the description of the present disclosure are omitted, and similar reference numerals are attached to similar parts.

[0043] In the present disclosure, when a component is “connected”, “coupled” or “linked” to another component, it may include not only a direct connection relationship but also an indirect connection relationship in which an intervening component is present. In addition, when a component “includes” or “has” other components, it means that other components may be further included, rather than excluding other components unless otherwise stated.

[0044] In the present disclosure, the terms first, second, etc. may be used only for the purpose of distinguishing one component from other components, and do not limit the order or importance of the components unless otherwise stated. Accordingly, within the scope of the present disclosure, a first component in one embodiment may be referred to as a second component in another embodiment, and similarly, a second component in one embodiment may be referred to as a first component in another embodiment.

[0045] In the present disclosure, components that are distinguished from each other are intended to clearly describe each feature, and do not mean that the components are necessarily separated. That is, a plurality of components may be integrated and implemented in one hardware or software unit, or one component may be distributed and implemented in a plurality of hardware or software units. Therefore, even if not stated otherwise, such embodiments in which the components are integrated or the component is distributed are also included in the scope of the present disclosure.

[0046] In the present disclosure, the components described in various embodiments do not necessarily mean essential components, and some components may be optional com-

ponents. Accordingly, an embodiment consisting of a subset of components described in an embodiment is also included in the scope of the present disclosure. In addition, embodiments including other components in addition to components described in the various embodiments are included in the scope of the present disclosure.

[0047] The present disclosure relates to encoding and decoding of an image, and terms used in the present disclosure may have a general meaning commonly used in the technical field, to which the present disclosure belongs, unless newly defined in the present disclosure.

[0048] In the present disclosure, a “picture” generally refers to a unit representing one image in a specific time period, and a slice/tile is a coding unit constituting a part of a picture, and one picture may be composed of one or more slices/tiles. In addition, a slice/tile may include one or more coding tree units (CTUs).

[0049] In the present disclosure, a “pixel” or a “pel” may mean a smallest unit constituting one picture (or image). In addition, “sample” may be used as a term corresponding to a pixel. A sample may generally represent a pixel or a value of a pixel, and may represent only a pixel/pixel value of a luma component or only a pixel/pixel value of a chroma component.

[0050] In the present disclosure, a “unit” may represent a basic unit of image processing. The unit may include at least one of a specific region of the picture and information related to the region. The unit may be used interchangeably with terms such as “sample array”, “block” or “area” in some cases. In a general case, an M×N block may include samples (or sample arrays) or a set (or array) of transform coefficients of M columns and N rows.

[0051] In the present disclosure, “current block” may mean one of “current coding block”, “current coding unit”, “coding target block”, “decoding target block” or “processing target block”. When prediction is performed, “current block” may mean “current prediction block” or “prediction target block”. When transform (inverse transform)/quantization (dequantization) is performed, “current block” may mean “current transform block” or “transform target block”. When filtering is performed, “current block” may mean “filtering target block”.

[0052] In addition, in the present disclosure, a “current block” may mean a block including both a luma component block and a chroma component block or “a luma block of a current block” unless explicitly stated as a chroma block. The luma component block of the current block may be expressed by including an explicit description of a luma component block such as “luma block” or “current luma block”. In addition, the “chroma component block of the current block” may be expressed by including an explicit description of a chroma component block, such as “chroma block” or “current chroma block”.

[0053] In the present disclosure, the term “/” and “,” should be interpreted to indicate “and/or.” For instance, the expression “A/B” and “A, B” may mean “A and/or B.” Further, “A/B/C” and “A/B/C” may mean “at least one of A, B, and/or C.”

[0054] In the present disclosure, the term “or” should be interpreted to indicate “and/or.” For instance, the expression “A or B” may comprise 1) only “A”, 2) only “B”, and/or 3) both “A and B”. In other words, in the present disclosure, the term “or” should be interpreted to indicate “additionally or alternatively.”

Overview of Video Coding System

[0055] FIG. 1 is a view illustrating a video coding system, to which an embodiment of the present disclosure is applicable.

[0056] The video coding system according to an embodiment may include an encoding apparatus 10 and a decoding apparatus 20. The encoding apparatus 10 may deliver encoded video and/or image information or data to the decoding apparatus 20 in the form of a file or streaming via a digital storage medium or network.

[0057] The encoding apparatus 10 according to an embodiment may include a video source generator 11, an encoding unit 12 and a transmitter 13. The decoding apparatus 20 according to an embodiment may include a receiver 21, a decoding unit 22 and a renderer 23. The encoding unit 12 may be called a video/image encoding unit, and the decoding unit 22 may be called a video/image decoding unit. The transmitter 13 may be included in the encoding unit 12. The receiver 21 may be included in the decoding unit 22. The renderer 23 may include a display and the display may be configured as a separate device or an external component.

[0058] The video source generator 11 may acquire a video/image through a process of capturing, synthesizing or generating the video/image. The video source generator 11 may include a video/image capture device and/or a video/image generating device. The video/image capture device may include, for example, one or more cameras, video/image archives including previously captured video/images, and the like. The video/image generating device may include, for example, computers, tablets and smartphones, and may (electronically) generate video/images. For example, a virtual video/image may be generated through a computer or the like. In this case, the video/image capturing process may be replaced by a process of generating related data.

[0059] The encoding unit 12 may encode an input video/image. The encoding unit 12 may perform a series of procedures such as prediction, transform, and quantization for compression and coding efficiency. The encoding unit 12 may output encoded data (encoded video/image information) in the form of a bitstream.

[0060] The transmitter 13 may transmit the encoded video/image information or data output in the form of a bitstream to the receiver 21 of the decoding apparatus 20 through a digital storage medium or a network in the form of a file or streaming. The digital storage medium may include various storage mediums such as USB, SD, CD, DVD, Blu-ray, HDD, SSD, and the like. The transmitter 13 may include an element for generating a media file through a predetermined file format and may include an element for transmission through a broadcast/communication network. The receiver 21 may extract/receive the bitstream from the storage medium or network and transmit the bitstream to the decoding unit 22.

[0061] The decoding unit 22 may decode the video/image by performing a series of procedures such as dequantization, inverse transform, and prediction corresponding to the operation of the encoding unit 12.

[0062] The renderer 23 may render the decoded video/image. The rendered video/image may be displayed through the display.

Overview of Image Encoding Apparatus

[0063] FIG. 2 is a view schematically illustrating an image encoding apparatus, to which an embodiment of the present disclosure is applicable.

[0064] As shown in FIG. 2, the image encoding apparatus **100** may include an image partitioner **110**, a subtractor **115**, a transformer **120**, a quantizer **130**, a dequantizer **140**, an inverse transformer **150**, an adder **155**, a filter **160**, a memory **170**, an inter prediction unit **180**, an intra prediction unit **185** and an entropy encoder **190**. The inter prediction unit **180** and the intra prediction unit **185** may be collectively referred to as a “predictor”. The transformer **120**, the quantizer **130**, the dequantizer **140** and the inverse transformer **150** may be included in a residual processor. The residual processor may further include the subtractor **115**.

[0065] All or at least some of the plurality of components configuring the image encoding apparatus **100** may be configured by one hardware component (e.g., an encoder or a processor) in some embodiments. In addition, the memory **170** may include a decoded picture buffer (DPB) and may be configured by a digital storage medium.

[0066] The image partitioner **110** may partition an input image (or a picture or a frame) input to the image encoding apparatus **100** into one or more processing units. For example, the processing unit may be called a coding unit (CU). The coding unit may be acquired by recursively partitioning a coding tree unit (CTU) or a largest coding unit (LCU) according to a quad-tree binary-tree ternary-tree (QT/BT/TT) structure. For example, one coding unit may be partitioned into a plurality of coding units of a deeper depth based on a quad tree structure, a binary tree structure, and/or a ternary structure. For partitioning of the coding unit, a quad tree structure may be applied first and the binary tree structure and/or ternary structure may be applied later. The coding procedure according to the present disclosure may be performed based on the final coding unit that is no longer partitioned. The largest coding unit may be used as the final coding unit or the coding unit of deeper depth acquired by partitioning the largest coding unit may be used as the final coding unit. Here, the coding procedure may include a procedure of prediction, transform, and reconstruction, which will be described later. As another example, the processing unit of the coding procedure may be a prediction unit (PU) or a transform unit (TU). The prediction unit and the transform unit may be split or partitioned from the final coding unit. The prediction unit may be a unit of sample prediction, and the transform unit may be a unit for deriving a transform coefficient and/or a unit for deriving a residual signal from the transform coefficient.

[0067] The prediction unit (the inter prediction unit **180** or the intra prediction unit **185**) may perform prediction on a block to be processed (current block) and generate a predicted block including prediction samples for the current block. The prediction unit may determine whether intra prediction or inter prediction is applied on a current block or CU basis. The prediction unit may generate various information related to prediction of the current block and transmit the generated information to the entropy encoder **190**. The information on the prediction may be encoded in the entropy encoder **190** and output in the form of a bitstream.

[0068] The intra prediction unit **185** may predict the current block by referring to the samples in the current picture. The referred samples may be located in the neighborhood of the current block or may be located apart

according to the intra prediction mode and/or the intra prediction technique. The intra prediction modes may include a plurality of non-directional modes and a plurality of directional modes. The non-directional mode may include, for example, a DC mode and a planar mode. The directional mode may include, for example, 33 directional prediction modes or 65 directional prediction modes according to the degree of detail of the prediction direction. However, this is merely an example, more or less directional prediction modes may be used depending on a setting. The intra prediction unit **185** may determine the prediction mode applied to the current block by using a prediction mode applied to a neighboring block.

[0069] The inter prediction unit **180** may derive a predicted block for the current block based on a reference block (reference sample array) specified by a motion vector on a reference picture. In this case, in order to reduce the amount of motion information transmitted in the inter prediction mode, the motion information may be predicted in units of blocks, subblocks, or samples based on correlation of motion information between the neighboring block and the current block. The motion information may include a motion vector and a reference picture index. The motion information may further include inter prediction direction (L0 prediction, L1 prediction, Bi prediction, etc.) information. In the case of inter prediction, the neighboring block may include a spatial neighboring block present in the current picture and a temporal neighboring block present in the reference picture. The reference picture including the reference block and the reference picture including the temporal neighboring block may be the same or different. The temporal neighboring block may be called a collocated reference block, a co-located CU (colCU), and the like. The reference picture including the temporal neighboring block may be called a collocated picture (colPic). For example, the inter prediction unit **180** may configure a motion information candidate list based on neighboring blocks and generate information indicating which candidate is used to derive a motion vector and/or a reference picture index of the current block. Inter prediction may be performed based on various prediction modes. For example, in the case of a skip mode and a merge mode, the inter prediction unit **180** may use motion information of the neighboring block as motion information of the current block. In the case of the skip mode, unlike the merge mode, the residual signal may not be transmitted. In the case of the motion vector prediction (MVP) mode, the motion vector of the neighboring block may be used as a motion vector predictor, and the motion vector of the current block may be signaled by encoding a motion vector difference and an indicator for a motion vector predictor. The motion vector difference may mean a difference between the motion vector of the current block and the motion vector predictor.

[0070] The prediction unit may generate a prediction signal based on various prediction methods and prediction techniques described below. For example, the prediction unit may not only apply intra prediction or inter prediction but also simultaneously apply both intra prediction and inter prediction, in order to predict the current block. A prediction method of simultaneously applying both intra prediction and inter prediction for prediction of the current block may be called combined inter and intra prediction (CIIP). In addition, the prediction unit may perform intra block copy (IBC) for prediction of the current block. Intra block copy may be

used for content image/video coding of a game or the like, for example, screen content coding (SCC). IBC is a method of predicting a current picture using a previously reconstructed reference block in the current picture at a location apart from the current block by a predetermined distance. When IBC is applied, the location of the reference block in the current picture may be encoded as a vector (block vector) corresponding to the predetermined distance. IBC basically performs prediction in the current picture, but may be performed similarly to inter prediction in that a reference block is derived within the current picture. That is, IBC may use at least one of the inter prediction techniques described in the present disclosure.

[0071] The prediction signal generated by the prediction unit may be used to generate a reconstructed signal or to generate a residual signal. The subtractor **115** may generate a residual signal (residual block or residual sample array) by subtracting the prediction signal (predicted block or prediction sample array) output from the prediction unit from the input image signal (original block or original sample array). The generated residual signal may be transmitted to the transformer **120**.

[0072] The transformer **120** may generate transform coefficients by applying a transform technique to the residual signal. For example, the transform technique may include at least one of a discrete cosine transform (DCT), a discrete sine transform (DST), a karhunen-loève transform (KLT), a graph-based transform (GBT), or a conditionally non-linear transform (CNT). Here, the GBT means transform obtained from a graph when relationship information between pixels is represented by the graph. The CNT refers to transform acquired based on a prediction signal generated using all previously reconstructed pixels. In addition, the transform process may be applied to square pixel blocks having the same size or may be applied to blocks having a variable size rather than square.

[0073] The quantizer **130** may quantize the transform coefficients and transmit them to the entropy encoder **190**. The entropy encoder **190** may encode the quantized signal (information on the quantized transform coefficients) and output a bitstream. The information on the quantized transform coefficients may be referred to as residual information. The quantizer **130** may rearrange quantized transform coefficients in a block type into a one-dimensional vector form based on a coefficient scanning order and generate information on the quantized transform coefficients based on the quantized transform coefficients in the one-dimensional vector form.

[0074] The entropy encoder **190** may perform various encoding methods such as, for example, exponential Golomb, context-adaptive variable length coding (CAVLC), context-adaptive binary arithmetic coding (CABAC), and the like. The entropy encoder **190** may encode information necessary for video/image reconstruction other than quantized transform coefficients (e.g., values of syntax elements, etc.) together or separately. Encoded information (e.g., encoded video/image information) may be transmitted or stored in units of network abstraction layers (NALs) in the form of a bitstream. The video/image information may further include information on various parameter sets such as an adaptation parameter set (APS), a picture parameter set (PPS), a sequence parameter set (SPS), or a video parameter set (VPS). In addition, the video/image information may further include general constraint information. The signaled

information, transmitted information and/or syntax elements described in the present disclosure may be encoded through the above-described encoding procedure and included in the bitstream.

[0075] The bitstream may be transmitted over a network or may be stored in a digital storage medium. The network may include a broadcasting network and/or a communication network, and the digital storage medium may include various storage media such as USB, SD, CD, DVD, Blu-ray, HDD, SSD, and the like. A transmitter (not shown) transmitting a signal output from the entropy encoder **190** and/or a storage unit (not shown) storing the signal may be included as internal/external element of the image encoding apparatus **100**. Alternatively, the transmitter may be provided as the component of the entropy encoder **190**.

[0076] The quantized transform coefficients output from the quantizer **130** may be used to generate a residual signal. For example, the residual signal (residual block or residual samples) may be reconstructed by applying dequantization and inverse transform to the quantized transform coefficients through the dequantizer **140** and the inverse transformer **150**.

[0077] The adder **155** adds the reconstructed residual signal to the prediction signal output from the inter prediction unit **180** or the intra prediction unit **185** to generate a reconstructed signal (reconstructed picture, reconstructed block, reconstructed sample array). If there is no residual for the block to be processed, such as a case where the skip mode is applied, the predicted block may be used as the reconstructed block. The adder **155** may be called a reconstructor or a reconstructed block generator. The generated reconstructed signal may be used for intra prediction of a next block to be processed in the current picture and may be used for inter prediction of a next picture through filtering as described below.

[0078] The filter **160** may improve subjective/objective image quality by applying filtering to the reconstructed signal. For example, the filter **160** may generate a modified reconstructed picture by applying various filtering methods to the reconstructed picture and store the modified reconstructed picture in the memory **170**, specifically, a DPB of the memory **170**. The various filtering methods may include, for example, deblocking filtering, a sample adaptive offset, an adaptive loop filter, a bilateral filter, and the like. The filter **160** may generate various information related to filtering and transmit the generated information to the entropy encoder **190** as described later in the description of each filtering method. The information related to filtering may be encoded by the entropy encoder **190** and output in the form of a bitstream.

[0079] The modified reconstructed picture transmitted to the memory **170** may be used as the reference picture in the inter prediction unit **180**. When inter prediction is applied through the image encoding apparatus **100**, prediction mismatch between the image encoding apparatus **100** and the image decoding apparatus may be avoided and encoding efficiency may be improved.

[0080] The DPB of the memory **170** may store the modified reconstructed picture for use as a reference picture in the inter prediction unit **180**. The memory **170** may store the motion information of the block from which the motion information in the current picture is derived (or encoded) and/or the motion information of the blocks in the picture that have already been reconstructed. The stored motion

information may be transmitted to the inter prediction unit **180** and used as the motion information of the spatial neighboring block or the motion information of the temporal neighboring block. The memory **170** may store reconstructed samples of reconstructed blocks in the current picture and may transfer the reconstructed samples to the intra prediction unit **185**.

Overview of Image Decoding Apparatus

[0081] FIG. 3 is a view schematically illustrating an image decoding apparatus, to which an embodiment of the present disclosure is applicable.

[0082] As shown in FIG. 3, the image decoding apparatus **200** may include an entropy decoder **210**, a dequantizer **220**, an inverse transformer **230**, an adder **235**, a filter **240**, a memory **250**, an inter prediction unit **260** and an intra prediction unit **265**. The inter prediction unit **260** and the intra prediction unit **265** may be collectively referred to as a “predictor”. The dequantizer **220** and the inverse transformer **230** may be included in a residual processor.

[0083] All or at least some of a plurality of components configuring the image decoding apparatus **200** may be configured by a hardware component (e.g., a decoder or a processor) according to an embodiment. In addition, the memory **170** may include a decoded picture buffer (DPB) or may be configured by a digital storage medium.

[0084] The image decoding apparatus **200**, which has received a bitstream including video/image information, may reconstruct an image by performing a process corresponding to a process performed by the image encoding apparatus **100** of FIG. 2. For example, the image decoding apparatus **200** may perform decoding using a processing unit applied in the image encoding apparatus. Thus, the processing unit of decoding may be a coding unit, for example. The coding unit may be acquired by partitioning a coding tree unit or a largest coding unit. The reconstructed image signal decoded and output through the image decoding apparatus **200** may be reproduced through a reproducing apparatus (not shown).

[0085] The image decoding apparatus **200** may receive a signal output from the image encoding apparatus of FIG. 2 in the form of a bitstream. The received signal may be decoded through the entropy decoder **210**. For example, the entropy decoder **210** may parse the bitstream to derive information (e.g., video/image information) necessary for image reconstruction (or picture reconstruction). The video/image information may further include information on various parameter sets such as an adaptation parameter set (APS), a picture parameter set (PPS), a sequence parameter set (SPS), or a video parameter set (VPS). In addition, the video/image information may further include general constraint information. The image decoding apparatus may further decode picture based on the information on the parameter set and/or the general constraint information. Signaled/received information and/or syntax elements described in the present disclosure may be decoded through the decoding procedure and obtained from the bitstream. For example, the entropy decoder **210** decodes the information in the bitstream based on a coding method such as exponential Golomb coding, CAVLC, or CABAC, and output values of syntax elements required for image reconstruction and quantized values of transform coefficients for residual. More specifically, the CABAC entropy decoding method may receive a bin corresponding to each syntax element in

the bitstream, determine a context model using a decoding target syntax element information, decoding information of a neighboring block and a decoding target block or information of a symbol/bin decoded in a previous stage, and perform arithmetic decoding on the bin by predicting a probability of occurrence of a bin according to the determined context model, and generate a symbol corresponding to the value of each syntax element. In this case, the CABAC entropy decoding method may update the context model by using the information of the decoded symbol/bin for a context model of a next symbol/bin after determining the context model. The information related to the prediction among the information decoded by the entropy decoder **210** may be provided to the prediction unit (the inter prediction unit **260** and the intra prediction unit **265**), and the residual value on which the entropy decoding was performed in the entropy decoder **210**, that is, the quantized transform coefficients and related parameter information, may be input to the dequantizer **220**. In addition, information on filtering among information decoded by the entropy decoder **210** may be provided to the filter **240**. Meanwhile, a receiver (not shown) for receiving a signal output from the image encoding apparatus may be further configured as an internal/external element of the image decoding apparatus **200**, or the receiver may be a component of the entropy decoder **210**.

[0086] Meanwhile, the image decoding apparatus according to the present disclosure may be referred to as a video/image/picture decoding apparatus. The image decoding apparatus may be classified into an information decoder (video/image/picture information decoder) and a sample decoder (video/image/picture sample decoder). The information decoder may include the entropy decoder **210**. The sample decoder may include at least one of the dequantizer **220**, the inverse transformer **230**, the adder **235**, the filter **240**, the memory **250**, the inter prediction unit **160** or the intra prediction unit **265**.

[0087] The dequantizer **220** may dequantize the quantized transform coefficients and output the transform coefficients. The dequantizer **220** may rearrange the quantized transform coefficients in the form of a two-dimensional block. In this case, the rearrangement may be performed based on the coefficient scanning order performed in the image encoding apparatus. The dequantizer **220** may perform dequantization on the quantized transform coefficients by using a quantization parameter (e.g., quantization step size information) and obtain transform coefficients.

[0088] The inverse transformer **230** may inversely transform the transform coefficients to obtain a residual signal (residual block, residual sample array).

[0089] The prediction unit may perform prediction on the current block and generate a predicted block including prediction samples for the current block. The prediction unit may determine whether intra prediction or inter prediction is applied to the current block based on the information on the prediction output from the entropy decoder **210** and may determine a specific intra/inter prediction mode (prediction technique).

[0090] It is the same as described in the prediction unit of the image encoding apparatus **100** that the prediction unit may generate the prediction signal based on various prediction methods (techniques) which will be described later.

[0091] The intra prediction unit **265** may predict the current block by referring to the samples in the current

picture. The description of the intra prediction unit **185** is equally applied to the intra prediction unit **265**.

[0092] The inter prediction unit **260** may derive a predicted block for the current block based on a reference block (reference sample array) specified by a motion vector on a reference picture. In this case, in order to reduce the amount of motion information transmitted in the inter prediction mode, motion information may be predicted in units of blocks, subblocks, or samples based on correlation of motion information between the neighboring block and the current block. The motion information may include a motion vector and a reference picture index. The motion information may further include inter prediction direction (L0 prediction, L1 prediction, Bi prediction, etc.) information. In the case of inter prediction, the neighboring block may include a spatial neighboring block present in the current picture and a temporal neighboring block present in the reference picture. For example, the inter prediction unit **260** may configure a motion information candidate list based on neighboring blocks and derive a motion vector of the current block and/or a reference picture index based on the received candidate selection information. Inter prediction may be performed based on various prediction modes, and the information on the prediction may include information indicating a mode of inter prediction for the current block.

[0093] The adder **235** may generate a reconstructed signal (reconstructed picture, reconstructed block, reconstructed sample array) by adding the obtained residual signal to the prediction signal (predicted block, predicted sample array) output from the prediction unit (including the inter prediction unit **260** and/or the intra prediction unit **265**). If there is no residual for the block to be processed, such as when the skip mode is applied, the predicted block may be used as the reconstructed block. The description of the adder **155** is equally applicable to the adder **235**. The adder **235** may be called a reconstructor or a reconstructed block generator. The generated reconstructed signal may be used for intra prediction of a next block to be processed in the current picture and may be used for inter prediction of a next picture through filtering as described below.

[0094] The filter **240** may improve subjective/objective image quality by applying filtering to the reconstructed signal. For example, the filter **240** may generate a modified reconstructed picture by applying various filtering methods to the reconstructed picture and store the modified reconstructed picture in the memory **250**, specifically, a DPB of the memory **250**. The various filtering methods may include, for example, deblocking filtering, a sample adaptive offset, an adaptive loop filter, a bilateral filter, and the like.

[0095] The (modified) reconstructed picture stored in the DPB of the memory **250** may be used as a reference picture in the inter prediction unit **260**. The memory **250** may store the motion information of the block from which the motion information in the current picture is derived (or decoded) and/or the motion information of the blocks in the picture that have already been reconstructed. The stored motion information may be transmitted to the inter prediction unit **260** so as to be utilized as the motion information of the spatial neighboring block or the motion information of the temporal neighboring block. The memory **250** may store reconstructed samples of reconstructed blocks in the current picture and transfer the reconstructed samples to the intra prediction unit **265**.

[0096] In the present disclosure, the embodiments described in the filter **160**, the inter prediction unit **180**, and the intra prediction unit **185** of the image encoding apparatus **100** may be equally or correspondingly applied to the filter **240**, the inter prediction unit **260**, and the intra prediction unit **265** of the image decoding apparatus **200**.

Example of Coding Layer Structure

[0097] A coded video/image according to the present disclosure may be processed, for example, according to a coding layer and structure which will be described below.

[0098] FIG. **4** is a view illustrating an example of a layer structure for a coded image/video.

[0099] The coded image/video is classified into a video coding layer (VCL) for an image/video decoding process and handling itself, a lower system for transmitting and storing encoded information, and a network abstraction layer (NAL) present between the VCL and the lower system and responsible for a network adaptation function.

[0100] In the VCL, VCL data including compressed image data (slice data) may be generated or a supplemental enhancement information (SEI) message additionally required for a decoding process of an image or a parameter set including information such as a picture parameter set (PPS), a sequence parameter set (SPS) or a video parameter set (VPS) may be generated.

[0101] In the NAL, header information (NAL unit header) may be added to a raw byte sequence payload (RBSP) generated in the VCL to generate a NAL unit. In this case, the RBSP refers to slice data, a parameter set, an SEI message generated in the VCL. The NAL unit header may include NAL unit type information specified according to RBSP data included in a corresponding NAL unit.

[0102] As shown in FIG. **4**, the NAL unit may be classified into a VCL NAL unit and a non-VCL NAL unit according to the type of the RBSP generated in the VCL. The VCL NAL unit may mean a NAL unit including information on an image (slice data), and the Non-VCL NAL unit may mean a NAL unit including information (parameter set or SEI message) required to decode an image.

[0103] The VCL NAL unit and the Non-VCL NAL unit may be attached with header information and transmitted through a network according to the data standard of the lower system. For example, the NAL unit may be modified into a data format of a predetermined standard, such as H.266/VVC file format, RTP (Real-time Transport Protocol) or TS (Transport Stream), and transmitted through various networks.

[0104] As described above, in the NAL unit, a NAL unit type may be specified according to the RBSP data structure included in the corresponding NAL unit, and information on the NAL unit type may be stored in a NAL unit header and signaled. For example, this may be largely classified into a VCL NAL unit type and a non-VCL NAL unit type depending on whether the NAL unit includes information on an image (slice data). The VCL NAL unit type may be classified according to the property and type of the picture included in the VCL NAL unit, and the Non-VCL NAL unit type may be classified according to the type of a parameter set.

[0105] An example of the NAL unit type specified according to the type of the parameter set/information included in the Non-VCL NAL unit type will be listed below.

- [0106] DCI (Decoding capability information) NAL unit type (NUT): Type for NAL unit including DCI
- [0107] VPS (Video Parameter Set) NAL unit: Type for NAL unit including VPS
- [0108] SPS (Sequence Parameter Set) NAL unit: Type for NAL unit including SPS
- [0109] PPS (Picture Parameter Set) NAL unit: Type for NAL unit including PPS
- [0110] APS (Adaptation Parameter Set) NAL unit: Type for NAL unit including APS
- [0111] PH (Picture header) NAL unit: Type for NAL unit including a picture header

[0112] The above-described NAL unit types may have syntax information for a NAL unit type, and the syntax information may be stored in a NAL unit header and signaled. For example, the syntax information may be `nal_unit_type`, and the NAL unit types may be specified using `nal_unit_type` values.

[0113] Meanwhile, one picture may include a plurality of slices, and one slice may include a slice header and slice data. In this case, one picture header may be further added to a plurality of slices (slice header and slice data set) in one picture. The picture header (picture header syntax) may include information/parameters commonly applicable to the picture. The slice header (slice header syntax) may include information/parameters commonly applicable to the slice. The APS (APS syntax) or PPS (PPS syntax) may include information/parameters commonly applicable to one or more slices or pictures. The SPS (SPS syntax) may include information/parameters commonly applicable to one or more sequences. The VPS (VPS syntax) may include information/parameters commonly applicable to multiple layers. The

DCI (DCI syntax) may include information/parameters related to decoding capability.

[0114] In the present disclosure, a high level syntax (HLS) may include at least one of the APS syntax, the PPS syntax, the SPS syntax, the VPS syntax, the DCI syntax, the picture header syntax or the slice header syntax. In addition, in the present disclosure, a low level syntax (LLS) may include, for example, a slice data syntax, a CTU syntax, a coding unit syntax, a transform unit syntax, etc.

[0115] In the present disclosure, image/video information encoded in the encoding apparatus and signaled to the decoding apparatus in the form of a bitstream may include not only in-picture partitioning related information, intra/inter prediction information, residual information, in-loop filtering information but also information on the slice header, information on the picture header, information on the APS, information on the PPS, information on the SPS, information on the VPS and/or information on the DCI. In addition, the image/video information may further include general constraint information and/or information on a NAL unit header.

Overview of Mixed NAL Unit Type

[0116] In general, one NAL unit type may be set for one picture. As described above, syntax information specifying a NAL unit type may be signaled in a NAL unit header of a NAL unit including a slice (data). Here, the syntax information may be `nal_unit_type` and NAL unit types may be specified using a `nal_unit_type` value.

[0117] An example of NAL unit types to which an embodiment of the present disclosure is applicable is shown in Table 1 below.

TABLE 1

nal_unit_type	Name of nal_unit_type	Content of NAL unit and RBSP syntax structure	NAL unit type class
0	TRAIL_NUT	Coded slice of a trailing picture slice_layer_rbsp()	VCL
1	STSA_NUT	Coded slice of an STSA picture slice_layer_rbsp()	VCL
2	RADL_NUT	Coded slice of a RADL picture slice_layer_rbsp()	VCL
3	RASL_NUT	Coded slice of a RASL picture slice_layer_rbsp()	VCL
4 . . . 6	RSV_VCL_4 . . . RSV_VCL_6	Reserved non-IRAP VCL NAL unit types	VCL
7	IDR_W_RADL	Coded slice of an IDR picture slice_layer_rbsp()	VCL
8	IDR_N_LP	Coded slice of a CRA picture silce_layer_rbsp()	VCL
9	CRA_NUT	Coded slice of a GDR picture slice_layer_rbsp()	VCL
10	GDR_NUT	Reserved IRAP VCL NAL unit types	VCL
11	RSV_IRAP_11		VCL
12	RSV_IRAP_12		VCL
13	DCI_NUT	Decoding capability information decoding_capability_information_rbsp()	non-VCL
14	VPS_NUT	Video parameter set video_parameter_set_rbsp()	non-VCL
15	SPS_NUT	Sequence parameter set seq_parameter_set_rbsp()	non-VCL
16	PPS_NUT	Picture parameter set pic_parameter_set_rbsp()	non-VCL
17	PREFIX_APS_NUT	Adaptation parameter set	non-VCL
18	SUFFIX_APS_NUT	adaptation_parameter_set_rbsp()	non-VCL
19	PH_NUT	Picture header picture_header_rbsp()	non-VCL

TABLE 1-continued

	Name of nal_unit_type	Content of NAL unit and RBSP syntax structure	NAL unit type class
20	AUD_NUT	AU delimiter access_unit_delimiter_rbsp()	non-VCL
21	EOS_NUT	End of sequence end_of_seq_rbsp()	non-VCL
22	EOB_NUT	End of bitstream end_of_bitstream_rbsp()	non-VCL
23	PREFIX_SEI_NUT	Supplemental enhancement information	non-VCL
24	SUFFIX_SEI_NUT	sei_rbsp()	non-VCL
25	FD_NUT	Filler data filler_data_rbsp()	non-VCL
26	RSV_NVCL_26	Reserved non-VCL NAL unit types	non-VCL
27	RSV_NVCL_27		non-VCL
28 . . . 31	UNSPEC_28 . . . UNSPEC 31	Unspecified non-VCL NAL unit types	non-VCL

NUT: NAL unit type

STSA: Step-wise Temporal sub-layer Switching Access

RADL: Random Access Decodable Leading

RASL: Random Access Slapped Leading

IDR: Instantaneous Decoding Refresh

LP: Leading Picture

_W_RADL: With RADL

_N_LP: No LP, without LP

CRA: Clean Random Access

GDR: Gradual Decoding Refresh

- IRAP: Intra Random Access Point

[0118] Referring to Table 1, a CRA picture may have associated RASL or RADL pictures present in a bitstream.

[0119] An IDR picture having nal_unit_type such as IDR_N_LP may not have associated leading pictures present in a bitstream. In contrast, an IDR picture having nal_unit_type such as IDR_W_RADL does not have associated RASL pictures present in a bitstream but may have associated RADL pictures present in the bitstream.

[0120] Meanwhile, the VVC standard allows slices in one picture having different NAL unit types. In this case, a syntax element pps_mixed_nalu_types_in_pic_flag in a picture parameter set (PPS) may have a first value (e.g., 1) and the following constraints may be applied.

[0121] The picture is constrained to include at least two subpictures.

[0122] VCL NAL units of the picture are constrained to have two or more different nal_unit_type values.

[0123] A VCL NAL unit having nal_unit_type such as GDR_NUT among VCL NAL units of the picture is constrained not to be present.

[0124] When VCL NAL units of at least one subpicture of the picture has nal_unit_type such as IDR_W_RADL, IDR_N_LP or CRA_NUT, all VCL NAL units of other subpictures in the picture are constrained to have nal_unit_type such as TRAIL_NUT.

[0125] In addition, when one picture contains mixed slices of RASL_NUT and RADL_NUT, the picture may be regarded as a RASL picture and a variable PictureOutputFlag specifying whether to output the picture may be derived as follows.

[0126] (1) When sps_video_parameter_set_id is greater than 0 and a current layer containing a current picture is not an output layer (i.e., for i in the range of 0 to NumOutputLayersInOls[TargetOlsIdx]-1, nuh_layer_id is not OutputLayerIdInOls[TargetOlsIdx][i]), PictureOutputFlag of the current picture may be set to a second value (e.g., 0). Alternatively, when any one of the following first and

second conditions is true, PictureOutputFlag of the current picture may be set to a second value (e.g., 0).

[0127] (First condition) a case where a current picture is a RASL picture and NoOutputBeforeRecoveryFlag of an associated IRAP picture has a first value (e.g., 1). Here, NoOutputBeforeRecoveryFlag of a first value (e.g., 1) may specify that pictures preceding a recovery point picture in decoding order may not be output before the recovery point picture.

[0128] (Second condition) a case where a current picture is a GDR picture having NoOutputBeforeRecoveryFlag of a first value (e.g., 1) or a recovering picture of a GDR picture having NoOutputBeforeRecoveryFlag of a first value (e.g., 1).

[0129] (2) In the other case, PictureOutputFlag may be set to the same value as ph_pic_output_flag.

[0130] In implementation, a decoder may output a picture which does not belong to an output layer. For example, while only one output layer is present in an access unit (AU), when the picture of the output layer is not available (for example, due to loss or layer down-switching), the decoder may set PictureOutputFlag to a first value (e.g., 1), for a picture having nuh_layer_id of a highest value among all pictures of the access unit (AU) available for the decoder and having ph_pic_output_flag of a first value (e.g., 1). In addition, in this case, the decoder may set PictureOutputFlag to a second value (e.g., 0) for all other pictures of the access unit (AU) available for the decoder.

Multi-Layer Based Coding

[0131] Image/video coding according to the present disclosure may include multi-layer based image/video coding. The multi-layer based image/video coding may include scalable coding. In multi-layer based coding or scalable coding, input signals may be processed for each layer. According to the layer, input signals (input images/videos) may have different values in terms of at least one of

resolution, frame rate, bit-depth, color format, aspect ratio or view. In this case, it is possible to reduce redundant information transmission/processing and to increase compression efficiency, by performing inter-layer prediction using a difference between layers (e.g., based on scalability).

[0132] FIG. 5 is a schematic block diagram of a multi-layer encoding apparatus 2100, to which embodiment(s) of the present disclosure is applicable, and, in which encoding of a multi-layer video/image signal is performed.

[0133] The multi-layer encoding apparatus 500 of FIG. 5 may include the encoding apparatus of FIG. 2. Compared to FIG. 2, the image partitioner 110 and the adder 155 are not shown in the multi-layer encoding apparatus 500 of FIG. 5, the multi-layer encoding apparatus 500 may include the image partitioner 110 and the adder 155. In this case, the image partitioner 110 and the adder 155 may be included in unit of layer. Hereinafter, multi-layer based prediction will be focused upon in the description of FIG. 5. For example, in addition to the following description, the multi-layer encoding apparatus 500 may include the technical idea of the encoding apparatus described above with reference to FIG. 2.

[0134] For convenience of description, a multi-layer structure consisting of two layers is shown in FIG. 5. However, the embodiments of the present disclosure are not limited to two layers and the multi-layer structure, to which the embodiment of the present disclosure is applied, may include two or more layers.

[0135] Referring to FIG. 5, the encoding apparatus 500 includes an encoder 500-1 of layer 1 and an encoder 500-0 of layer 0. Layer 0 may be a base layer, a reference layer or a lower layer, and layer 1 may be an enhancement layer, a current layer or a higher layer.

[0136] The encoder 500-1 of layer 1 may include a predictor 520-1, a residual processor 530-1, a filter 560-1, a memory 570-1, an entropy encoder 540-1 and a multiplexer (MUX) 570-. In an embodiment, the MUX may be included as an external component.

[0137] The encoder 500-0 of layer 0 may include a predictor 520-0, a residual processor 530-0, a filter 560-0, a memory 570-0 and an entropy encoder 540-0.

[0138] The predictors 520-0 and 520-1 may perform prediction with respect to input images based on various prediction schemes as described above. For example, the predictors 520-0 and 520-1 may perform inter prediction and intra prediction. The predictors 520-0 and 520-1 may perform prediction in a predetermined processing unit. The prediction unit may be a coding unit (CU) or a transform unit (TU). A predicted block (including prediction samples) may be generated according to the result of prediction and, based on this, the residual processor may derive a residual block (including residual samples).

[0139] Through inter prediction, prediction may be performed based on information on at least one of a previous picture and/or a next picture of a current picture, thereby generating a prediction block. Through intra prediction, prediction may be performed based on neighboring samples in a current picture, thereby generating a prediction block.

[0140] As an inter prediction mode or method, the above-described various prediction modes or methods may be used. In inter prediction, a reference picture may be selected for a current block to be predicted, and a reference block corresponding to the current block may be selected from the

reference picture. The predictors 520-0 and 520-1 may generate a predicted block based on the reference block.

[0141] In addition, the predictor 520-1 may perform prediction for layer 1 using information on layer 0. In the present disclosure, a method of predicting information on a current layer using information on another layer is referred to as inter-layer prediction, for convenience of description.

[0142] Information on a current layer predicted using information on another layer (that is, predicted by inter-layer prediction) may be at least one of texture, motion information, unit information or a predetermined parameter (e.g., a filtering parameter, etc.).

[0143] In addition, information on another layer used for prediction of the current layer (that is, used for inter-layer prediction) may be at least one of texture, motion information, unit information or a predetermined parameter (e.g., a filtering parameter, etc.).

[0144] Inter-layer prediction, a current block may be a block in a current picture in a current layer (e.g., layer 1) and may be a block to be coded. A reference block is a block in a picture (reference picture) belonging to the same access unit (AU) as a picture (current picture), to which the current block belongs, on a layer (reference layer, e.g., layer 0) referred to for prediction of the current block, and may be a block corresponding to the current block.

[0145] As an example of inter-layer prediction, there is inter-layer motion prediction for predicting motion information of a current layer using motion information of a reference layer. According to inter-layer motion prediction, motion information of a current block may be predicted using motion information of a reference block. That is, in deriving motion information according to the inter prediction mode which will be described below, a motion information candidate may be derived based on motion information of an inter-layer reference block instead of a temporal neighboring block.

[0146] When inter-layer motion prediction is applied, the predictor 520-1 may scale and use reference block (that is, inter-layer reference block) motion information of the reference layer.

[0147] As another example of inter-layer prediction, inter-layer texture prediction may use texture of a reconstructed reference block as a prediction value for a current block. In this case, the predictor 520-1 may scale the texture of the reference block by up-scaling. Inter-layer texture prediction may be referred to as inter-layer (reconstructed) sample prediction or simply inter-layer prediction.

[0148] In inter-layer parameter prediction which is another example of inter-layer prediction, a derived parameter of a reference layer may be reused in a current layer or a parameter for a current layer may be derived based on a parameter used in a reference layer.

[0149] In inter-layer residual prediction which is another example of inter-layer prediction, residual information of a current layer may be predicted using residual information of another layer and, based on this, prediction of a current block may be performed.

[0150] In inter-layer difference prediction which is another example of inter-layer prediction, prediction of a current block may be performed using a difference between images obtained by up-sampling or down-sampling a reconstructed picture of a current layer and a reconstructed picture of a reference layer.

[0151] In inter-layer syntax prediction which is another example of inter-layer prediction, texture of a current block may be predicted or generated using syntax information of a reference layer. In this case, syntax information of a referenced reference layer may include information on an intra prediction mode and motion information.

[0152] A plurality of prediction methods using the above-described inter-layer may be used when predicting a particular block.

[0153] Here, as an example of inter-layer prediction, although inter-layer texture prediction, inter-layer motion prediction, inter-layer unit information prediction, inter-layer parameter prediction, inter-layer residual prediction, inter-layer difference prediction, inter-layer syntax prediction, etc. are described, inter-layer prediction applicable in the present disclosure is not limited thereto.

[0154] For example, inter-layer prediction may be applied as an extension of inter prediction for a current layer. That is, by including a reference picture derived from a reference layer in reference pictures capable of being referenced for inter prediction of a current block, it is possible to perform inter prediction for the current block.

[0155] In this case, the inter-layer reference picture may be included in a reference picture list for the current block. The predictor 520-1 may perform inter prediction for the current block using an inter-layer reference picture.

[0156] Here, the inter-layer reference picture may be a reference picture constructed by sampling the reconstructed picture of a reference layer to correspond to the current layer. Accordingly, when the reconstructed picture of the reference layer corresponds to the picture of the current layer, the reconstructed picture of the reference layer may be used as an inter-layer reference picture without sampling. For example, when the widths and heights of samples are the same in the reconstructed picture of the reference layer and the reconstructed picture of the current layer and an offset between the top left end, top right end, bottom left end and bottom right end in the picture of the reference layer and the top left end, top right end, bottom left end and bottom right end in the picture of the current layer is 0, the reconstructed picture of the reference layer may be used as the inter-layer reference picture of the current layer without being sampled again.

[0157] In addition, the reconstructed picture of the reference layer, from which the inter-layer reference picture is derived, may be a picture belonging to the same AU as the current picture to be encoded.

[0158] When inter prediction for a current block is performed by including an inter-layer reference picture in a reference picture list, the position of the inter-layer reference picture in the reference picture list may be different between reference picture lists L0 and L1. For example, in reference picture list L0, the inter-layer reference picture may be located after short-term reference pictures before the current picture and, in reference picture list L1, the inter-layer reference picture may be located at the end of the reference picture list.

[0159] Here, reference picture list L0 is a reference picture list used for inter prediction of P slice or a reference picture list used as a first reference picture list in inter prediction of B slice. Reference picture list L1 may be a second reference picture list used for inter prediction of B slice.

[0160] Accordingly, reference picture list L0 may consist of short-term reference picture(s) before the current picture,

an inter-layer reference picture, short-term reference picture(s) after the current picture and a long-term reference picture in this order. Reference picture list L1 may consist of short-term reference picture(s) after the current picture, short-term reference picture(s) before the current picture, a long-term reference picture and an inter-layer reference picture in this order.

[0161] In this case, a predictive (P) slice is a slice for which intra prediction is performed or inter prediction is performed using a maximum of one motion vector per prediction block and a reference picture index. A bi-predictive (B) slice is a slice for which intra prediction is performed or prediction is performed using a maximum of two motion vectors per prediction block and a reference picture index. In this regard, an intra (I) slice is a slice to which only intra prediction is applied.

[0162] In addition, when inter prediction for a current block is performed based on a reference picture list including an inter-layer reference picture, the reference picture list may include a plurality of inter-layer reference pictures derived from a plurality of layers.

[0163] When a plurality of inter-layer reference pictures is included, the inter-layer reference pictures may be alternately arranged in reference picture lists L0 and L1. For example, assume that two inter-layer reference pictures such as inter-layer reference picture ILRP_i and inter-layer reference picture ILRP_j are included in the reference picture list used for inter prediction of the current block. In this case, in reference picture list L0, ILRP_i may be located after short-term reference pictures before the current picture and ILRP_j may be located at the end of the list. In addition, in reference picture list L1, ILRP_i may be located at the end of the list and ILRP_j may be located after short-term reference pictures after the current picture.

[0164] In this case, reference picture list L0 may consist of short-term reference picture(s) before the current picture, an inter-layer reference picture ILRP_i, short-term reference picture(s) after the current picture, a long-term reference picture and an inter-layer reference picture ILRP_j in this order. Reference picture list L1 may consist of short-term reference picture(s) after the current picture, an inter-layer reference picture ILRP_j, short-term reference picture(s) before the current picture, a long-term reference picture and an inter-layer reference picture ILRP_i in this order.

[0165] In addition, one of two inter-layer reference pictures may be an inter-layer reference picture derived from a scalable layer for resolution and the other may be an inter-layer reference picture derived from a layer for providing another view. In this case, for example, if ILRP_i is an inter-layer reference picture derived from a layer for providing different resolution and ILRP_j is an inter-layer reference picture derived from a layer for providing a different view, in the case of scalable video coding supporting only scalability excluding a view, reference picture list L0 may consist of short-term reference picture(s) before the current picture, an inter-layer reference picture ILRP_i, short-term reference picture(s) after the current picture, and a long-term reference picture in this order, and reference picture list L1 may consist of short-term reference picture(s) after the current picture, short-term reference picture(s) before the current picture, a long-term reference picture and an inter-layer reference picture ILRP_i in this order.

[0166] Meanwhile, in inter-layer prediction, as information on an inter-layer reference picture, only a sample value

may be used, only motion information (motion vector) may be used or both a sample value and motion information may be used. The predictor **520-1** may use only a sample value of the inter-layer reference picture, may use only motion information (motion vector) of the inter-layer reference picture or may use both the sample value and motion information of the inter-layer reference picture according to information received from the encoding apparatus, when the reference picture index indicates an inter-layer reference picture.

[0167] When only the sample value of the inter-layer reference picture is used, the predictor **220-1** may derive samples of a block specified by a motion vector from the inter-layer reference picture as a prediction sample of a current block. In the case of scalable video coding which does not take a view into consideration, a motion vector in inter prediction (inter-layer prediction) using an inter-layer reference picture may be set to a fixed value (e.g., 0).

[0168] When only the motion information of the inter-layer reference picture is used, the predictor **520-1** may use a motion vector specified by the inter-layer reference picture as a motion vector predictor for deriving the motion vector of the current block. In addition, the predictor **520-1** may use a motion vector specified by the inter-layer reference picture as the motion vector of the current block.

[0169] When both the sample value and motion information of the inter-layer reference picture are used, the predictor **520-1** may use, for prediction of the current block, a sample of a region corresponding to the current block in the inter-layer reference picture and motion information (motion vector) specified in the inter-layer reference picture.

[0170] The encoding apparatus may transmit a reference index indicating an inter-layer reference picture in a reference picture list to the decoding apparatus, when inter-layer prediction is applied, and may transmit, to the decoding apparatus, information for specifying which information (sample information, motion information or sample information and motion information) is used from the inter-layer reference picture, that is, information for specifying a dependency type of dependency for inter-layer prediction between two layers.

[0171] FIG. 6 is a schematic block diagram of a decoding apparatus, to which embodiment(s) of the present disclosure is applicable, and, in which decoding of a multi-layer video/image signal is performed. The decoding apparatus of FIG. 6 may include the decoding apparatus of FIG. 3. A realigner shown in FIG. 6 may be omitted or included in a dequantizer. In the description of this drawing, multi-layer based prediction will be focused upon. In addition, the description of the decoding apparatus of FIG. 3 may be included.

[0172] In the example of FIG. 6, for convenience of description, a multi-layer structure consisting of two layers will be described. However, it should be noted that embodiments of the present disclosure are not limited thereto, and the multi-layer structure, to which the embodiment of the present disclosure is applied, may include two or more layers.

[0173] Referring to FIG. 6, the decoding apparatus **600** may include a decoder **600-1** of layer 1 and a decoder **600-0** of layer 0. The decoder **600-1** of layer 1 may include an entropy decoder **610-1**, a residual processor **620-1**, a predictor **630-1**, an adder **640-1**, a filter **650-1** and a memory **660-1**. The decoder **600-2** of layer 0 may include an entropy

decoder **610-0**, a residual processor **620-0**, a predictor **630-0**, an adder **640-0**, a filter **650-0** and a memory **660-0**.

[0174] When a bitstream including image information is received from the encoding apparatus, a DEMUX **605** may demultiplex information for each layer and transmit the information to the decoding apparatus for each layer.

[0175] The entropy decoders **610-1** and **610-0** may perform decoding in correspondence with a coding method used in the encoding apparatus. For example, when CABAC is used in the encoding apparatus, the entropy decoders **610-1** and **610-0** may perform entropy decoding using CABAC.

[0176] When a prediction mode for a current block is an intra prediction mode, the predictors **630-1** and **630-0** may perform intra prediction for the current block based on neighboring reconstructed samples in the current picture.

[0177] When an prediction mode for a current block is an inter prediction mode, the predictors **630-1** and **630-0** may perform inter prediction for the current block based on information included in at least one of picture before or after the current picture. Some or all of motion information necessary for inter prediction may be derived by checking information received from the encoding apparatus.

[0178] When a skip mode is applied as an inter prediction mode, residual is not transmitted from the encoding apparatus and a prediction block may be a reconstructed block.

[0179] Meanwhile, the predictor **630-1** of layer 1 may perform inter prediction or intra prediction using only information on layer 1 and perform inter-layer prediction using information on another layer (layer 0).

[0180] As information on a current layer predicted using information on another layer (e.g., predicted by inter-layer prediction), there may be at least one of texture, motion information, unit information, a predetermined parameter (e.g., a filtering parameter, etc.).

[0181] As information on another layer used for prediction of the current layer (e.g., used for inter-layer prediction), there may be at least one of texture, motion information, unit information, a predetermined parameter (e.g., a filtering parameter, etc.).

[0182] In inter-layer prediction, a current block may be a block in a current picture in a current layer (e.g., layer 1) and may be a block to be decoded. A reference block may be a block in a picture (reference picture) belonging to the same access unit (AU) as a picture (current picture), to which the current block belongs, on a layer (reference layer, e.g., layer 0) referenced for prediction of the current block and may be a block correspond to the current block.

[0183] The multi-layer decoding apparatus **600** may perform inter-layer prediction as described in the multi-layer encoding apparatus **500**. For example, the multi-layer decoding apparatus **600** may perform inter-layer texture prediction, inter-layer motion prediction, inter-layer unit information prediction, inter-layer parameter prediction, inter-layer residual prediction, inter-layer difference prediction, inter-layer syntax prediction, etc., as described in the multi-layer encoding apparatus **500** and inter-layer prediction applicable in the present disclosure is not limited thereto.

[0184] The predictor **630-1** may perform inter-layer prediction using an inter-layer reference picture, when a reference picture index received from the encoding apparatus or a reference picture index derived from a neighboring block indicates an inter-layer reference picture in a reference picture list. For example, the predictor **630-1** may derive a sample value of a region specified by a motion vector in the

inter-layer reference picture as a prediction block for a current block, when a reference picture index indicates an inter-layer reference picture.

[0185] In this case, the inter-layer reference picture may be included in a reference picture list for a current block. The predictor 630-1 may perform inter prediction for the current block using the inter-layer reference picture.

[0186] As described above in the multi-layer encoding apparatus 500, in operation of the multi-layer decoding apparatus 600, the inter-layer reference picture may be a reference picture constructed by sampling a reconstructed picture of a reference layer to correspond to a current layer. Processing for the case where the reconstructed picture of the reference layer corresponds to the picture of the current layer may be performed in the same manner as the encoding process.

[0187] In addition, as described above in the multi-layer encoding apparatus 500, in operation of the multi-layer decoding apparatus 600, a reconstructed picture of a reference layer, from which an inter-layer reference picture is derived, may be a picture belonging to the same AU as a current picture to be encoded.

[0188] In addition, as described above in the multi-layer encoding apparatus 500, in operation of the multi-layer decoding apparatus 600, when inter prediction for a current block is performed by including the inter-layer reference picture in a reference picture list, the position of the inter-layer reference picture in the reference picture list may be different between reference picture lists L0 and L1.

[0189] In addition, as described above in the multi-layer encoding apparatus 500, in operation of the multi-layer decoding apparatus 600, when inter prediction for a current block is performed based on a reference picture list including an inter-layer reference picture, the reference picture list may include a plurality of inter-layer reference pictures derived from a plurality of layers, and arrangement of inter-layer reference pictures may be performed to correspond to that described in the encoding process.

[0190] In addition, as described above in the multi-layer encoding apparatus 500, in operation of the multi-layer decoding apparatus 600, as information on an inter-layer reference picture, only a sample value may be used, only motion information (motion vector) may be used or both a sample value and motion information may be used.

[0191] The multi-layer decoding apparatus 600 may receive a reference index indicating an inter-layer reference picture in a reference picture list from the multi-layer encoding apparatus 500 and perform inter-layer prediction based on the same. In addition, the multi-layer decoding apparatus 600 may receive, from the multi-layer encoding apparatus 500, information for specifying which information (sample information, motion information or sample information and motion information) is used from the inter-layer reference picture, that is, information for specifying a dependency type of dependency for inter-layer prediction between two layers.

[0192] An image encoding method and decoding method performed by a multi-layer image encoding apparatus and a multi-layer image decoding apparatus according to an embodiment will be described with reference to FIGS. 7 and 8. Hereinafter, for convenience of description, the multi-layer image encoding apparatus may be referred to as an

image encoding apparatus. In addition, the multi-layer image decoding apparatus may be referred to as an image decoding apparatus.

[0193] FIG. 7 is a view illustrating a method of encoding an image based on a multi-layer structure by an image encoding apparatus according to an embodiment. The image encoding apparatus according to the embodiment may encode picture(s) of a first layer (S710). Next, the image encoding apparatus may encode picture(s) of a second layer based on the first layer (S720). Next, the image encoding apparatus may output a bitstream (for a multi-layer) (S730).

[0194] FIG. 8 is a view illustrating a method of decoding an image based on a multi-layer structure by an image decoding apparatus according to an embodiment. The image decoding apparatus according to the embodiment may obtain video/image information from a bitstream (S810). Next, the image decoding apparatus may decode picture(s) of a first layer based on the video/image information (S820). Next, the image decoding apparatus may decode picture(s) of a second layer based on the video/image information and the first layer (S830).

[0195] In an embodiment, the video/image information may include high level syntax (HLS) described below. In an embodiment, the HLS may include an SPS and/or a PPS disclosed in the present disclosure. For example, the video/image information may include information and/or syntax elements described in the present disclosure. As described in the present disclosure, the picture(s) of the second layer may be encoded based on motion information/reconstructed samples/parameters of the picture(s) of the first layer. In an embodiment, the layer may be a layer lower than the second layer. In an embodiment, when the second layer is a current layer, the first layer may be referred to as a reference layer.

Poc (Picture Order Count) Decoding Process

[0196] A process for deriving a POC value for a current picture is as follows.

[0197] Output of the process may be PicOrderCntVal specifying a POC value of the current picture.

[0198] Each coded picture may be associated with a POC variable referred to as PicOrderCntVal.

[0199] When `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` has a second value (e.g., 0) and a picture picA is present in a current access unit (AU) in a reference layer of a current layer, PicOrderCntVal of the current picture may be derived to be equal to PicOrderCntVal of the picture picA, and the value of `ph_order_cnt_lsb` may be forced to be the same in all VCL NAL units of the current access unit (AU). Alternatively, PicOrderCntVal of the current picture may be derived as follows.

[0200] It is assumed that `prevTid0Pic` has `nuh_layer_id` equal to `nuh_layer_id` of the current picture, has `TemporalId` and `ph_non_ref_pic_flag` equal to 0, and is a previous picture in decoding order, not a RASL or RADL picture. Here, `ph_non_ref_pic_flag` may specify whether the current picture is able to be used as a reference picture. For example, `ph_non_ref_pic_flag` of a first value (e.g., 1) may specify that the current picture is never used as a reference picture. In contrast, `ph_non_ref_pic_flag` of a second value (e.g., 0) may specify that the current picture is able to be used as a reference picture.

[0201] A variable `prevPicOrderCntLsb` is set equal to `ph_pic_order_cnt_lsb` of `prevTid0Pic`.

- [0202] A variable `prevPicOrderCntMsb` is set equal to `PicOrderCntMsb` of `prevTid0Pic`.
- [0203] In this case, a variable `PicOrderCntMsb` of the current picture may be derived as follows.
- [0204] When `ph_poc_msb_cycle_present_flag` has a first value (e.g., 1), `PicOrderCntMsb` may be set equal to `h_poc_msb_cycle_val*MaxPicOrderCntLsb`.
- [0205] In contrast, when `ph_poc_msb_cycle_present_flag` has a second value (e.g., 0) and the current picture is a CLVSS picture, `PicOrderCntMsb` may be set to 0.
- [0206] In other cases, `PicOrderCntMsb` may be derived as shown in Table 2 below.

TABLE 2

<pre> if((ph_pic_order_cnt_lsb < prevPicOrderCntLsb) && ((prevPicOrderCntLsb - ph_pic_order_cnt_lsb) >= (MaxPicOrderCntLsb / 2))) PicOrderCntMsb = prevPicOrderCntMsb + MaxPicOrderCntLsb else if((ph_pic_order_cnt_lsb > prevPicOrderCntLsb) && ((ph_pic_order_cnt_lsb - prevPicOrderCntLsb) > (MaxPicOrderCntLsb / 2))) PicOrderCntMsb = prevPicOrderCntMsb - MaxPicOrderCntLsb else PicOrderCntMsb = prevPicOrderCntMsb </pre>

[0207] Referring to Table 2, when `prevPicOrderCntLsb` is greater than or equal to a value obtained by adding `MaxPicOrderCntLsb/2` to `ph_pic_order_cnt_lsb`, `PicOrderCntMsb` may be derived to be a value obtained by adding `MaxPicOrderCntLsb` to `prevPicOrderCntMsb`. In `n` `prevPicOrderCntLsb` is less than a value obtained by adding `MaxPicOrderCntLsb/2` to `ph_pic_order_cnt_lsb`, `PicOrderCntMsb` may be derived to be a value obtained by subtracting `MaxPicOrderCntLsb` from `prevPicOrderCntMsb`. In the other case, `PicOrderCntMsb` may be derived to be the same value as `prevPicOrderCntMsb`.

[0208] By adding `ph_pic_order_cnt_lsb` to `PicOrderCntMsb` derived in this way, `PicOrderCntVal` may be derived (i.e., `PicOrderCntVal=PicOrderCntMsb+ph_pic_order_cnt_lsb`). In this case, all CLVSS pictures in which `ph_poc_msb_cycle_val` is not present may have the same `PicOrderCntVal` as `ph_pic_order_cnt_lsb`, because `PicOrderCntMsb` is set to 0 for the pictures.

[0209] In an example, the value of `PicOrderCntVal` may be constrained in a range from -2^{31} to $2^{31}-1$.

[0210] In an example, in one CVS, for any two coded picture having the same `nuh_layer_id` value, the values of `PicOrderCntVal` may be constrained not to be equal to each other.

[0211] In an example, all pictures in any specific access unit (AU) may be constrained to have `PicOrderCntVal` of the same value.

[0212] In an example, a function `PicOrderCnt(picX)` may specify `PicOrderCntVal` of a picture `picX` (i.e., `PicOrderCnt(picX)=PicOrderCntVal` of the picture `picX`).

[0213] In an example, a function `DiffPicOrderCnt(picA, picB)` may specify a POC difference between a picture `picA` and a picture `picB` (i.e., `DiffPicOrderCnt(picA, picB)=PicOrderCnt(picA)-PicOrderCnt(picB)`). In this case, a bit-stream may be constrained such that a value of `DiffPicOrderCnt(picA, picB)` used in a decoding process does not contain data out of a range of -2^{15} to $2^{15}-1$. Meanwhile, assuming that a current picture is `X` and two different pictures in the same CVS are `Y` and `Z`, when both `DiffPi-`

`cOrderCnt(X, Y)` and `DiffPicOrderCnt(X, Z)` are positive or negative, `Y` and `Z` may be regarded as being in the same output order direction from `X`.

[0214] Hereinafter, problems which may occur during POC derivation under the multi-layer structure will be described.

Problems of Related Art

[0215] FIG. 9 is a view illustrating problems which may occur during POC derivation under a multi-layer structure.

[0216] Referring to FIG. 9, the multi-layer structure may consist of two layers of Layer 0 and Layer 1. Layer 0 is a lower layer and may be referred to as a base layer or a reference layer. In contrast, Layer 1 is a higher layer and may be referred to as an enhancement layer or a current layer. Hereinafter, it is assumed that Layer 0 is a non-output layer and Layer 1 is an output layer.

[0217] Layer 0 may include a plurality of Intra Random Access Point (IRAP) pictures having a single picture type. For example, Layer 0 may include first to `n`-th clean random access (CRA) pictures. In this case, within Layer 0, the first to `n`-th CRA pictures may be spaced apart from each other in picture order count (POC) order. For example, the POC value of the first CRA picture may be 0, the POC value of the second CRA picture may be 32, the POC value of the third CRA picture may be 64 and the POC value of the fourth CRA picture may be 96.

[0218] In contrast, Layer 1 may include a plurality of pictures having two or more picture types. For example, Layer 1 may include at least one CRA picture and a plurality of trail pictures. In this case, within Layer 1, the at least one CRA picture and the plurality of trail pictures may be sequentially arranged in POC order. For example, the POC value of the at least one CRA picture may be 0 and the POC values of the plurality of trail pictures may be greater than or equal to 1.

[0219] It is assumed that a value of a syntax element `sps_log2_max_pic_order_cnt_lsb_minus4` signaled within high level syntax (HLS), for example, a sequence parameter set (SPS) is 0 and `sps_poc_msb_cycle_flag` has a second value (e.g., 0). Here, `sps_log2_max_pic_order_cnt_lsb_minus4` may specify a maximum value (e.g., `MaxPicOrderCntLsb`) of a POC least significant bit (LSB) used for POC derivation. When the value of `sps_log2_max_pic_order_cnt_lsb_minus4` is 0, since only 4 bits are used for signaling of the POC LSB, a value of `MaxPicOrderCntLsb` may be 16. In addition, `sps_poc_msb_cycle_flag` may specify whether flag (e.g., `ph_poc_msb_cycle_present_flag`) information specifying whether POC most significant bit (MSB) cycle information (e.g., `ph_poc_msb_cycle_val`) used for POC derivation is present in a picture header is present in the picture header. When `sps_poc_msb_cycle_flag` has a second value (e.g., 0), `ph_poc_msb_cycle_present_flag` may not be present in the picture header.

[0220] Under the above-described conditions, when the second CRA picture, the third CRA picture or the fourth CRA picture in Layer 0 of FIG. 9 are decoded, since a decoder cannot find a previous picture (i.e., `prevTid0Pic`) in decoding order, a current POC may not be derived correctly. Accordingly, in the above-described example, in order to derive the current POC, POC MSB information (e.g., `ph_poc_msb_cycle_val`) for CRA pictures in Layer needs to be signaled. However, according to the existing POC deri-

vation process, there is a problem that POC MSB information is not forced to be signaled.

[0221] In order to solve such a problem, according to embodiments of the present disclosure, POC MSB information may be forced to be signaled under a predetermined condition. In addition, according to embodiments of the present disclosure, when POC MSB information is not signaled for a current picture, an extraction process may be modified such that a pre-specified previous picture (i.e., prevTid0Pic) is forced to be present or pictures with TemporalId equal to 0 is not removed.

[0222] The embodiments of the present disclosure may be implemented by modifying semantics of the above-described sps_poc_msb_cycle_flag and/or ph_poc_cycle_present_flag. SPS syntax containing sps_poc_msb_cycle_flag and picture header syntax containing ph_poc_cycle_present_flag will be briefly described and then embodiments of the present disclosure will be described.

[0223] FIG. 10 is a view illustrating an example of SPS syntax.

[0224] Referring to FIG. 10, an SPS may include syntax element sps_seq_parameter_set_id. sps_seq_parameter_set_id may provide an identifier for the SPS to be referred to by other syntax elements.

[0225] In addition, the SPS may include a syntax element sps_video_parameter_set_id. sps_video_parameter_set_id may specify a value of vps_video_parameter_set_id which is an identifier for a VPS referred to be by the SPS. Presence of a VPS may be optional for a single layer bitstream. When the VPS is not present, sps_video_parameter_set_id may be 0, and the following may be applied.

[0226] The SPS may not refer to the VPS, and, when each coded layer video sequence (CLVS) referring to the SPS is decoded, any VPS may not be referred to.

[0227] A value of vps_max_layers_minus1 may be inferred to be 0. Here, a value obtained by adding 1 to vps_max_layers_minus1 may specify the number of layers specified by the VPS, that is, a maximum allowed number of layers in each coded video sequence (CVS) referring to the VPS.

[0228] The value of vps_max_sublayers_minus1 may be inferred to be 6. Here, a value obtained by adding 1 to vps_max_sublayers_minus1 may specify a maximum number of temporal sublayers which may be present in a layer specified by the VPS. In an example, the value of vps_max_sublayers_minus1 may be constrained within a range of 0 to 6.

[0229] The CVS may be constrained to include only one layer. That is, all VCL NAL units in the CVS may be constrained to have nuh_layer_id of the same value.

[0230] The value of GeneralLayerIdx[nuh_layer_id] may be inferred to be 0.

[0231] The value of vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]] may be inferred to be 1. Here, vps_independent_layer_flag[i] may specify whether a layer having an index i is able to use inter layer prediction. For example, vps_independent_layer_flag[i] of a first value (e.g., 1) may specify that the layer having the index does not use inter layer prediction. In contrast, vps_independent_layer_flag[i] of a second value (e.g., 0) may specify that the layer having the index i is able to use inter layer prediction.

[0232] In addition, the SPS may include a syntax element sps_poc_msb_cycle_flag. sps_poc_msb_cycle_flag may

specify whether ph_poc_msb_cycle_present_flag is present in a picture header referring to the SPS. For example, sps_poc_msb_cycle_flag of a first value (e.g., 1) may specify that ph_poc_msb_cycle_present_flag is present in the picture header referring to the SPS. In contrast, sps_poc_msb_cycle_flag of a second value (e.g., 0) may specify that ph_poc_msb_cycle_present_flag is not present in the picture header referring to the SPS.

[0233] FIG. 11 is a view illustrating an example of picture header syntax.

[0234] Referring to FIG. 11, a picture header may include a syntax element ph_recovery_poc_cnt. ph_recovery_poc_cnt may specify a recovery point of decoded pictures in output order. In an example, when a current picture is a GDR picture and the value of ph_recovery_poc_cnt for current picture is 0, the current picture itself may be a recovery point picture.

[0235] In addition, a picture header may include a syntax element ph_poc_msb_cycle_present_flag. ph_poc_msb_cycle_present_flag may be encoded/decoded only when sps_poc_msb_cycle_flag described above with reference to FIG. 10 has a first value (e.g., 1). ph_poc_msb_cycle_present_flag may specify whether a syntax element ph_poc_msb_cycle_val is present in the picture header. For example, ph_poc_msb_cycle_present_flag of a first value (e.g., 1) may specify that the syntax element ph_poc_msb_cycle_val is present in the picture header. In contrast, ph_poc_msb_cycle_present_flag of a second value (e.g., 0) may specify that the syntax element ph_poc_msb_cycle_val is not present in the picture header. When vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]] described with reference to FIG. 10 has a second value (e.g., 0) and a picture is present in a current access unit (AU) in the reference layer of the current layer, the value of ph_poc_msb_cycle_present_flag may be constrained to a second value (e.g., 0).

[0236] In addition, the picture header may have a syntax element ph_poc_msb_cycle_val. ph_poc_msb_cycle_val may specify a POC MSB cycle value of the current picture. In an example, a bit length of ph_poc_msb_cycle_val may be equal to a value obtained by adding 1 to a syntax element sps_poc_msb_cycle_minus1 signaled in the SPS.

[0237] Hereinafter, embodiments of the present disclosure will be described in detail.

Embodiment 1 to 5

[0238] According to Embodiment 1 to Embodiment 5 of the present disclosure, for an IRAP picture and/or a GDR picture with an associated ph_recovery_poc_cnt value equal to 0, signaling of POC MSB information may be forced under a predetermined condition.

[0239] Specifically, according to Embodiment 1 of the present disclosure, for IRAP pictures in a layer including only IRAP pictures, not an output layer, as an independent layer, it may be constrained such that signaling of POC MSB information is present.

[0240] According to Embodiment 2 of the present disclosure, as IRAP pictures included in a layer which is not an output layer as an independent layer, when only the IRAP pictures are required for inter layer prediction by other layer(s), it may be constrained such that signaling of POC MSB information for the IRAP pictures is present.

[0241] According to Embodiment 3 of the present disclosure, as an independent layer, as IRAP pictures included in a layer which is not an output layer in at least one output

layer set (OLS) specified in a video parameter set (VPS), when only the IRAP pictures are required for inter layer prediction by other layer(s), it is constrained such that signaling of POC MSB information for the IRAP pictures is present.

[0242] According to Embodiment 4 of the present disclosure, constraints of Embodiment 1 to Embodiment 3 described above is applicable not only to IRAP pictures but also to a GDR (Gradual Decoding Refresh) picture with an associated `ph_recovery_poc_cnt` value equal to 0. Here, when the value of `ph_recovery_poc_cnt` associated with the GDR picture is 0, the GDR picture itself may be a recovery point picture.

[0243] According to Embodiment 5 of the present disclosure, constraints of Embodiment 1 to Embodiment 4 described above are applicable regardless of whether the above-described layer is an output layer. For example, when the layer of Embodiment 1 described above includes only IRAP pictures as an independent layer, it may be constrained such that signaling of POC MSB information for the IRAP pictures is present regardless of whether the layer is an output layer.

[0244] Embodiment 1 to Embodiment 5 described above may be implemented by modifying semantics of each of `sps_poc_msb_cycle_flag` and `ph_poc_cycle_present_flag`. Detailed implementations are as follows.

(1) Implementation 1

[0245] According to Implementation 1 of the present disclosure, Constraint 1 or 2 for bitstream conformance may be added to the semantics of `sps_poc_msb_cycle_flag` described with reference to FIG. 10 as follows.

[0246] (Constraint 1) When all the following conditions are true, `sps_poc_msb_cycle_flag` may be forced to have a first value (e.g., 1).

[0247] a case where the value of `sps_video_parameter_set_id` is not 0.

[0248] a case where an SPS is referred to by at least one VCL NAL unit of a layer having a layer id equal to `nuh_layer_id`, the value of `vps_max_tid_il_ref_pics_plus1[i][GeneralLayerIdx[nuh_layer_id]]` is 0 for all *i* values in a range of 0 to `vps_max_layers_minus1`, and the value of `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` is 0. Here, `vps_max_tid_il_ref_pics_plus1[i][j]` having a value of 0 may specify that pictures of a *j*-th layer which is neither IRAP pictures nor GDR pictures with `ph_recovery_poc_cnt` of 0 are not used as inter-layer reference pictures (ILRPs) in decoding pictures of an *i*-th layer. In contrast, `vps_max_tid_il_ref_pics_plus1[i][j]` having a value greater than 0 may specify that no picture of a *j*-th layer having `TemporalId` greater than `vps_max_tid_il_ref_pics_plus1[i][j]-1` is used as an inter-layer reference picture in decoding pictures of an *i*-th layer, and no adaptation parameter set (APS) having `TemporalId` greater than `vps_max_tid_il_ref_pics_plus1[i][j]-1` and `nuh_layer_id` equal to `vps_layer_id[j]` is not referred to.

[0249] (Constraint 2) Alternatively, when all the following conditions are true, `sps_poc_msb_cycle_flag` may be forced to have a first value (e.g., 1).

[0250] A case where the value of `sps_video_parameter_set_id` is not 0.

[0251] A case where the SPS is referred to by at least one VCL NAL unit of a layer having a layer id equal

to `nuh_layer_id` and, for all *i* values in a range of 0 to `vps_max_layers_minus1`, the value of `vps_max_tid_il_ref_pics_plus1[i][GeneralLayerIdx[nuh_layer_id]]` is 0.

[0252] In addition, according to Implementation 1 of the present disclosure, Constraint 3 or Constraint 4 for bitstream conformance may be added to semantics of `ph_poc_msb_cycle_present_flag` described above with reference to FIG. 11 as follows.

[0253] (Constraint 3) When all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0254] A case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0255] a case where the value of `sps_video_parameter_set_id` is not 0.

[0256] a case where, for all *i* values in a range of 0 to `vps_max_layers_minus1`, the value of `vps_max_tid_il_ref_pics_plus1[i][GeneralLayerIdx[nuh_layer_id]]` is 0.

[0257] (Constraint 4) Alternatively, when all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0258] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0259] a case where the value of `sps_video_parameter_set_id` is not 0.

[0260] For all *i* values in a range of 0 to `vps_max_layers_minus1`, the value of `vps_max_tid_il_ref_pics_plus1[i][GeneralLayerIdx[nuh_layer_id]]` is 0 and the value of `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` is 0.

(2) Implementation 2

[0261] According to Implementation 2 of the present disclosure, Constraint 5 or 6 for bitstream conformance may be added to the semantics of `sps_poc_msb_cycle_flag` described with reference to FIG. 10 as follows.

[0262] (Constraint 5) When all the following conditions are true, `sps_poc_msb_cycle_flag` may be forced to have a first value (e.g., 1).

[0263] a case where the value of `sps_video_parameter_set_id` is not 0.

[0264] a case where an SPS is referred to by at least one VCL NAL unit of a layer having a layer id equal to `nuh_layer_id`, and, for all *i* values in a range of 0 to `TotalNumOls-1`, the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` is 0. Here, `NumSubLayersInLayerInOLS[i][j]` may specify the number of sublayers in a *j*-th output layer in an *i*-th output layer set (OLS).

[0265] (Constraint 6) Alternatively, when all the following conditions are true, `sps_poc_msb_cycle_flag` may be forced to have a first value (e.g., 1).

[0266] a case where the value of `sps_video_parameter_set_id` is not 0.

[0267] A case where the SPS is referred to by at least one VCL NAL unit of a layer having a layer id equal to `nuh_layer_id`, the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` is 0 for

all i values in a range of 0 to $\text{TotalNumOlss}-1$, and the value of $\text{vps_independent_layer_flag}[\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0.

[0268] In addition, according to Implementation 2 of the present disclosure, Constraint 7 or Constraint 8 for bitstream conformance may be added to semantics of $\text{ph_poc_msb_cycle_present_flag}$ described above with reference to FIG. 11 as follows.

[0269] (Constraint 7) When all the following conditions are true, $\text{ph_poc_msb_cycle_present_flag}$ may be forced to have a first value (e.g., 1).

[0270] a case where a picture associated with a picture header including $\text{ph_poc_msb_cycle_present_flag}$ is an IRAP picture or a GDR picture with an associated $\text{ph_recovery_poc_cnt}$ value equal to 0.

[0271] a case where the value of $\text{sps_video_parameter_set_id}$ is not 0.

[0272] a case where, for all i value in a range of 0 to $\text{TotalNumOlss}-1$, the value of $\text{NumSubLayersInLayerInOLS}[i][\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0.

[0273] (Constraint 8) Alternatively, when all the following conditions are true, $\text{ph_poc_msb_cycle_present_flag}$ may be forced to have a first value (e.g., 1).

[0274] a case where a picture associated with a picture header including $\text{ph_poc_msb_cycle_present_flag}$ is an IRAP picture or a GDR picture with an associated $\text{ph_recovery_poc_cnt}$ value equal to 0.

[0275] a case where the value of $\text{sps_video_parameter_set_id}$ is not 0.

[0276] a case where, for all i values in a range of 0 to $\text{TotalNumOlss}-1$, the value of $\text{NumSubLayersInLayerInOLS}[i][\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0 and the value of $\text{vps_independent_layer_flag}[\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0.

(3) Implementation 3

[0277] According to Implementation 3 of the present disclosure, Constraint 9 or 10 for bitstream conformance may be added to the semantics of $\text{sps_poc_msb_cycle_flag}$ described with reference to FIG. 10 as follows.

[0278] (Constraint 9) When all the following conditions are true, $\text{sps_poc_msb_cycle_flag}$ may be forced to have a first value (e.g., 1).

[0279] a case where the value of $\text{sps_video_parameter_set_id}$ is not 0.

[0280] a case where an SPS is referred to by at least one VCL NAL unit of a layer having a layer id equal to nuh_layer_id and the value of $\text{NumSubLayersInLayerInOLS}[i][\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0.

[0281] (Constraint 10) Alternatively, when all the following conditions are true, $\text{sps_poc_msb_cycle_flag}$ may be forced to have a first value (e.g., 1).

[0282] a case where the value of $\text{sps_video_parameter_set_id}$ is not 0.

[0283] A case where the SPS is referred to by at least one VCL NAL unit of a layer having a layer id equal to nuh_layer_id , the value of $\text{NumSubLayersInLayerInOLS}[i][\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0, and the value of $\text{vps_independent_layer_flag}[\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0.

[0284] In addition, according to Implementation 3 of the present disclosure, Constraint 11 or Constraint 12 for bit-

stream conformance may be added to semantics of $\text{ph_poc_msb_cycle_present_flag}$ described above with reference to FIG. 11 as follows.

[0285] (Constraint 11) When all the following conditions are true, $\text{ph_poc_msb_cycle_present_flag}$ may be forced to have a first value (e.g., 1).

[0286] a case where a picture associated with a picture header including $\text{ph_poc_msb_cycle_present_flag}$ is an IRAP picture or a GDR picture with an associated $\text{ph_recovery_poc_cnt}$ value equal to 0.

[0287] a case where the value of $\text{sps_video_parameter_set_id}$ is not 0.

[0288] a case where the value of $\text{NumSubLayersInLayerInOLS}[i][\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0.

[0289] (Constraint 12) Alternatively, when all the following conditions are true, $\text{ph_poc_msb_cycle_present_flag}$ may be forced to have a first value (e.g., 1).

[0290] a case where a picture associated picture header including $\text{ph_poc_msb_cycle_present_flag}$ is an IRAP picture or a GDR picture with an associated $\text{ph_recovery_poc_cnt}$ value equal to 0.

[0291] a case where the value of $\text{sps_video_parameter_set_id}$ is not 0.

[0292] a case where the value of $\text{NumSubLayersInLayerInOLS}[i][\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0 and the value of $\text{vps_independent_layer_flag}[\text{GeneralLayerIdx}[\text{nuh_layer_id}]]$ is 0.

[0293] As described above, according to Embodiment 1 to Embodiment 5 of the present disclosure, for an IRAP picture and/or a GDR picture with an associated $\text{ph_recovery_poc_cnt}$ value equal to 0, signaling of POC MSB information may be forced under a predetermined condition. Therefore, when a non-output layer includes only the IRAP picture and/or the GDR picture (see FIG. 9), as a decoder cannot find a previous picture (i.e., prevTid0Pic) of the IRAP picture and/or the GDR picture in decoding order, the problem that the POC of the IRAP picture and/or the GDR picture is not derived correctly may be solved.

Embodiment 6

[0294] According to Embodiment 6 of the present disclosure, when there is no POC MSB information for an IRAP picture (and a GDR picture with an associated $\text{ph_recovery_poc_cnt}$ value equal to 0), it may be constrained that a non-output layer cannot be extracted such that only IRAP pictures (and a GDR picture with an associated $\text{ph_recovery_poc_cnt}$ value equal to 0) is present in the non-output layer after extraction. Here, when the value of $\text{ph_recovery_poc_cnt}$ associated with the GDR picture is 0, the GDR picture itself may be a recovery point picture.

[0295] In an embodiment, only when the current layer is a non-output layer and $\text{sps_poc_msb_cycle_flag}$ has a second value (e.g., 0), it may be constrained that the current layer cannot be extracted such that only IRAP pictures (and a GDR picture with an associated $\text{ph_recovery_poc_cnt}$ value equal to 0) are present in the current layer.

[0296] Embodiment 6 described above may be implemented by modifying the semantics of $\text{sps_poc_msb_cycle_flag}$. Specific implementations are as follows.

(1) Implementation 4

[0297] According to Implementation 4 of the present disclosure, Constraint 13 or 14 for bitstream conformance

may be added to the semantics of `sps_poc_msb_cycle_flag` described with reference to FIG. 10 as follows.

[0298] (Constraint 13) When all the following conditions are true, for any layer having a layer id equal to `nuh_layer_id`, the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` may be forced to be greater than 0, for an `i` value in a range 0 to `TotalNumOls-1`.

[0299] a case where, within a referenced SPS, the value of `sps_video_parameter_set_id` is not 0.

[0300] a case where, in the referenced SPS, `sps_poc_msb_cycle_flag` has a second value (e.g., 0).

[0301] a case where `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` has a second value (e.g., 0).

[0302] a case where the SPS is referred to by at least one VCL NAL unit of a layer having a layer id equal to `nuh_layer_id`, the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` is 0.

[0303] (Constraint 14) Alternatively, when all the following conditions are true, for any layer having a layer id equal to `nuh_layer_id`, the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` may be forced to be greater than 0, for an `i` value in a range 0 to `vps_max_layers_minus1`.

[0304] a case where, within the referenced SPS, the value of `sps_video_parameter_set_id` is not 0.

[0305] a case where, in the referenced SPS, `sps_poc_msb_cycle_flag` has a second value (e.g., 0).

[0306] a case where `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` has a second value (e.g., 0).

[0307] A case where the SPS is referred to by at least one VCL NAL unit of a layer having a layer id equal to `nuh_layer_id` and the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` is 0.

[0308] As described above, according to Embodiment 6 of the present disclosure, when there is no POC MSB information for an IRAP picture (and a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0), it may be constrained that a non-output layer cannot be extracted such that only IRAP pictures (and a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0) are present in the non-output layer after extraction. Therefore, when the non-output layer includes only IRAP pictures (and a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0) (see FIG. 9), as a decoder cannot find a previous picture (i.e., `prevTid0Pic`) of the IRAP picture (and the GDR picture) in decoding order, the problem that the POC of the IRAP picture (and the GDR picture) is not derived correctly may be solved.

Embodiment 7

[0309] According to Embodiment 7 of the present disclosure, when a current picture is not a CLVSS (coded layer video sequence start) picture (that is, the current picture is not a first picture of a CLVS (coded layer video sequence)) and there is no POC MSB information signaled for the current picture (i.e., `ph_poc_msb_cycle_present_flag=0`), it may be constrained that there is a picture specified as a previous picture (i.e., `prevTid0Pic`) of the current picture in decoding order. In addition, in this case, the specified picture may be constrained to be the same as the picture specified by an encoder for the current picture.

[0310] Embodiment 7 described above may be implemented by modifying the POC decoding process described above with reference to Table 2. Specific implementations are as follows.

(1) Implementation 5

[0311] According to Implementation 5 of the present disclosure, the POC decoding process described above with reference to Table 2 may be modified as follows.

[0312] When `ph_poc_msb_cycle_present_flag` has a second value (e.g., 0) and a current picture is not a CLVSS (coded layer video sequence start) picture, a variable `prevPicOrderCntLsb` and a variable `prevPicOrderCntMsb` may be derived as follows.

[0313] It is assumed that `prevTid0Pic` has `nuh_layer_id` equal to `nuh_layer_id` of the current picture, has `TemporalId` and `ph_non_ref_pic_flag` equal to 0, and is a previous picture in decoding order, not a RASL or RADL picture.

[0314] `prevTid0Pic` is constrained to be present and a POC difference between `prevTid0Pic` and the current picture is constrained to be less than `MaxPicOrderCntLsb/2`.

[0315] A variable `prevPicOrderCntLsb` is set equal to `ph_pic_order_cnt_lsb` of `prevTid0Pic`.

[0316] A variable `prevPicOrderCntMsb` is set equal to `PicOrderCntMsb` of `prevTid0Pic`.

[0317] As described above, according to Embodiment 7 of the present disclosure, when a current picture is not a CLVSS picture and there is no POC MSB information signaled for the current picture, it may be constrained that there is a picture specified as a previous picture (i.e., `prevTid0Pic`) of the current picture in decoding order, and the specified picture may be constrained to be equal to a picture specified by an encoder. Therefore, when a non-output layer includes only an IRAP picture and/or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0 (see FIG. 9), as a decoder cannot find a previous picture (i.e., `prevTid0Pic`) of the IRAP picture and/or the GDR picture in decoding order, the problem that the POC of the IRAP picture and/or the GDR picture is not derived correctly may be solved.

Embodiment 8

[0318] According to Embodiment 8 of the present disclosure, under certain conditions, an extraction process may be modified such that pictures with `TemporalId` equal to 0 in a layer is not removed. The predetermined conditions are as follows.

[0319] (Condition 1) a case where the layer is an independent layer.

[0320] (Condition 2) a case where the layer is not an output layer.

[0321] (Condition 3) a case where there is no POC MSB information for pictures in the layer, or there is no POC MSB information for IRAP pictures in the layer or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0. Here, when the value of `ph_recovery_poc_cnt` associated with the GDR picture is 0, the GDR picture itself may be recovery point picture.

[0322] Therefore, since pictures having `TemporalId` equal to 0 that are not removed in the extraction process may be used as a previous picture (i.e., `prevTid0Pic`) of the current

picture, as in the example of FIG. 9, the problem that the POC of the current picture is not derived correctly may be solved.

Embodiment 9

[0323] According to Embodiment 9 of the present disclosure, as IRAP pictures included in a layer which is not an output layer as an independent layer or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0, when only the IRAP pictures and the GDR picture are required for inter layer prediction by other layer(s) and a distance between the IRAP pictures or the GDR picture and a previous IRAP picture or GDR picture is greater than or equal to $\text{MaxPicOrderCntLsb}/2$, it may be constrained such that signaling of POC MSB information for the IRAP pictures or the GDR picture is present. Here, when the value of `ph_recovery_poc_cnt` associated with the GDR picture is 0, the GDR picture itself may be recovery point picture.

[0324] Embodiment 9 described above may be implemented by modifying the semantics of each of `sps_poc_msb_cycle_flag` and `ph_poc_cycle_present_flag`. To this end, Implementation 1 to Implementation 3 described above may be modified and applied.

(1) Modified Implementation 1

[0325] Specifically, in order to implement Embodiment 9, Constraint 3 and Constraint 4 of Implementation 1 described above may be modified as follows.

[0326] (Modified constraint 3) When all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0327] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0328] a case where the picture associated with the picture header is not a CLVSS picture.

[0329] a case where the value of `sps_video_parameter_set_id` is not 0.

[0330] a case where the value of `vps_max_tid_il_ref_pics_plus1[i][GeneralLayerIdx[nuh_layer_id]]` is 0 for all *i* value in a range of 0 to `vps_max_layers_minus1`.

[0331] (Modified constraint 4) Alternatively, when all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0332] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0333] a case where a picture associated with the picture header is not a CLVSS picture.

[0334] a case where the value of `sps_video_parameter_set_id` is not 0.

[0335] a case where, for all *i* values in a range of 0 to `vps_max_layers_minus1`, the value of `vps_max_tid_il_ref_pics_plus1[i][GeneralLayerIdx[nuh_layer_id]]` is 0 and the value of `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` is 0.

(2) Modified Implementation 2

[0336] In addition, in order to implement Embodiment 9, Constraint 7 and Constraint 8 of Implementation 2 described above may be modified as follows.

[0337] (Modified constraint 7) When all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0338] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0339] a case where a picture associated with the picture header is not a CLVSS picture.

[0340] a case where the value of `sps_video_parameter_set_id` is not 0.

[0341] a case where, for all *i* values in a range of 0 to `TotalNumOlss-1`, the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` is 0.

[0342] (Modified constraint 8) Alternatively, when all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0343] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0344] a case where a picture associated with the picture header is not a CLVSS picture.

[0345] a case where the value of `sps_video_parameter_set_id` is not 0.

[0346] a case where, for all *i* values in a range of 0 to `TotalNumOlss-1`, the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` is 0 and the value of `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` is 0.

(3) Modified Implementation 3

[0347] In addition, in order to implement Embodiment 9, Constraint 11 and Constraint 12 of Implementation 3 described above may be modified as follows.

[0348] (Modified constraint 11) When all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0349] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0350] a case where a picture associated with the picture header is not a CLVSS picture.

[0351] a case where the value of `sps_video_parameter_set_id` is not 0.

[0352] a case where the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` is 0.

[0353] (Modified constraint 12) Alternatively, when all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0354] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0355] a case where the picture associated with the picture header is not a CLVSS picture.

[0356] a case where the value of `sps_video_parameter_set_id` is not 0.

[0357] a case where the value of `NumSubLayersInLayerInOLS[i][GeneralLayerIdx[nuh_layer_id]]` is 0 and the value of `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` is 0.

[0358] Embodiment 9 described above may be implemented by modifying the semantics of `ph_poc_cycle_present_flag`. Specific implementations are as follows.

(4) Implementation 6

[0359] According to Implementation 6 of the present disclosure, Constraint 15 to Constraint 17 below may be added to the semantics of `ph_poc_msb_cycle_present_flag` described above with reference to FIG. 11.

[0360] (Constraint 15) When all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0361] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0362] a case where the picture is not a CLVSS picture.

[0363] a case where a POC difference between the picture and an IRAP picture immediately preceding the picture in decoding order or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0 is less than or equal to `MaxPicOrderCntLsb/2`.

[0364] a case where the value of `sps_video_parameter_set_id` is not 0 and the value of `NumSubLayersInLayerInOLS[TargetOlsIdx][GeneralLayerIdx[nuh_layer_id]]` is 0.

[0365] (Constraint 16) Alternatively, when all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0366] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0367] a case where the picture is not a CLVSS picture.

[0368] a case where a POC difference between the picture and an IRAP picture immediately preceding the picture in decoding order or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0 is less than or equal to `MaxPicOrderCntLsb/2`.

[0369] a case where the value of `sps_video_parameter_set_id` is not 0, the value of `NumSubLayersInLayerInOLS[TargetOlsIdx][GeneralLayerIdx[nuh_layer_id]]` is 0 and `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` has a first value (e.g., 1).

[0370] (Constraint 17) Alternatively, when all the following conditions are true, `ph_poc_msb_cycle_present_flag` may be forced to have a first value (e.g., 1).

[0371] a case where a picture associated with a picture header including `ph_poc_msb_cycle_present_flag` is an IRAP picture or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0.

[0372] a case where the picture is not a CLVSS picture.

[0373] a case where a POC difference between the picture and an IRAP picture immediately preceding the picture in decoding order or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0 is less than or equal to `MaxPicOrderCntLsb/2`.

[0374] a case where the value of `sps_video_parameter_set_id` is not 0, the value of `NumSubLayersInLayerInOLS[TargetOlsIdx][GeneralLayerIdx[nuh_layer_id]]` is 0 for an *i* value in 0 to `TotalNulOls` and a range of `-1`, `vps_independent_layer_flag[GeneralLayerIdx[nuh_layer_id]]` has a first value (e.g., 1).

[0375] Embodiment 9 described above may be implemented by modifying the POC decoding process described above with reference to Table 2. Specific implementations are as follows.

(5) Implementation 7

[0376] According to Implementation 7 of the present disclosure, the POC decoding process described with reference to Table 2 may be modified as follows.

[0377] When `ph_poc_msb_cycle_present_flag` has a second value (e.g., 0) and a current picture is not a CLVSS (coded layer video sequence start) picture, a variable `prevPicOrderCntLsb` and a variable `prevPicOrderCntMsb` may be derived as follows.

[0378] It is assumed that `prevTid0Pic` has `nuh_layer_id` equal to `nuh_layer_id` of the current picture, has `TemporalId` and `ph_non_ref_pic_flag` equal to 0, and is a previous picture in decoding order, not a RASL or RADL picture.

[0379] `prevTid0Pic` is constrained to be present, and a POC difference between `prevTid0Pic` and the current picture is constrained to be less than `MaxPicOrderCntLsb/2`.

[0380] A variable `prevPicOrderCntLsb` may be set equal to `ph_pic_order_cnt_lsb` of `prevTid0Pic`.

[0381] A variable `prevPicOrderCntMsb` is set equal to `PicOrderCntMsb` of `prevTid0Pic`.

[0382] As described above, according to Embodiment 9 of the present disclosure, for an IRAP picture and/or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0, signaling of POC MSB information may be forced under a predetermined condition. In this case, a distance between the IRAP picture and/or the GDR and a previous IRAP picture or GDR picture may be less than or equal to `MaxPicOrderCntLsb/2`. Therefore, when a non-output layer includes only the IRAP picture and/or the GDR picture (see FIG. 9), as a decoder cannot find a previous picture (i.e., `prevTid0Pic`) of the IRAP picture and/or the GDR picture in decoding order, the problem that the POC of the IRAP picture and/or the GDR picture is not derived correctly may be solved.

[0383] Hereinafter, an image encoding/decoding method according to an embodiment of the present disclosure will be described in detail with reference to the accompanying drawings.

[0384] FIG. 12 is a flowchart illustrating an image encoding method according to an embodiment of the present disclosure.

[0385] The image encoding method of FIG. 12 may be performed by the image encoding apparatus of FIG. 2 or 5. For example, step S1210 to step S1230 may be performed by the prediction unit 180, 520-0 or 520-1 and the entropy encoder 190, 540-0 or 540-1.

[0386] Referring to FIG. 12, the image encoding apparatus may determine a POC (picture order count) MSB (Most Significant Bit) of a current picture (S1210).

[0387] In an example, the POC MSB may be determined based on whether first information specifying a value of the POC MSB cycle of the current picture is present. For example, when the first information (e.g., `ph_poc_msb_cycle_val`) is present, the POC MSB may be derived by multiplying the value of the POC MSB cycle by a maximum value (e.g., `MaxPicOrderCntLsb`) of a POC LSB (Least Significant Bit) of the current picture. In contrast, when the

first information is not present and the current picture is a CLVSS (coded layer video sequence start) picture, the POC MSB may be derived to be 0. In the other case, the POC MSB may be derived based on a POC MSB (e.g., `prevPicOrderCntMsb`) of a previous picture preceding the current picture in decoding order, as described above with reference to Table 2.

[0388] The image encoding apparatus may derive the POC of the current picture based on the POC MSB determined in step S1210 (S1220). In an example, the POC (e.g., `PicOrderCntMsb`) may be derived by adding the POC LSB (e.g., `ph_pic_order_cnt_lsb`) to the POC MSB (e.g., `PicOrderCntMsb`).

[0389] The image encoding apparatus may encode information on the POC (S1230).

[0390] The information on the POC may include flag information (e.g., `sps_poc_msb_cycle_flag` and `ph_poc_msb_cycle_present_flag`) specifying whether first information specifying the value of the POC MSB cycle is present. In addition, the information on the POC may further include first information (e.g., `ph_poc_msb_cycle_val`) specifying the value of the POC MSB cycle, based on the first information being present (e.g., `sps_poc_msb_cycle_flag=1` && `ph_poc_msb_cycle_present_flag=1`).

[0391] In an example, based on the information on the POC doing not include the first information and the current picture being not a CLVSS (coded layer video sequence start) picture, a POC difference between the current picture and a previous picture (e.g., `Tid0Pic`) preceding the current picture in decoding order may be less than half a maximum value (e.g., `MaxPicOrderCntLsb`) of a POC LSB (Least Significant Bit) of the current picture. In this case, a temporal identifier (e.g., `TemporalId`) of the previous picture may be 0. In addition, a layer identifier (e.g., `nuh_layer_id`) for a NAL unit of the previous picture may be equal to a layer identifier value for a NAL unit of the current picture, and the previous picture may be used as a reference picture (e.g., `ph_non_ref_pic_flag=0`) and may not be a RASL (Random Access Skipped Leading) picture or a RADL (Random Access Decodable Leading) picture. The above-described embodiment may be defined as in Implementation 7.

[0392] In an embodiment, based on the current picture being an IRAP (intra random access point) picture or a GDR picture (i.e., a GDR picture with associated `ph_recovery_poc_cnt` equal to 0) as a recovery point picture, the first information may be encoded in a picture header. In this case, the IRAP picture or the GDR picture may not be a CLVSS (coded layer video sequence start) picture in an independent non-output layer. In addition, only IRAP pictures and a GDR picture in a current layer including the current picture may be used for inter layer prediction by layers different from the current layer. In addition, a distance between the current picture and a previous IRAP picture preceding the current picture in decoding order or a GDR picture as a recovery point picture may be greater than or equal to half a maximum value (e.g., `MaxPicOrderCntLsb`) of the POC LSB.

[0393] The above-described embodiment may be defined as a constraint for bitstream conformance as in Embodiment 9 of the present disclosure. That is, as IRAP pictures included in a layer which is not an output layer as an independent layer or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0, only the IRAP pictures and the GDR picture are required for inter layer

prediction by other layer(s), and, when a distance between the IRAP pictures or the GDR picture and a previous IRAP picture or GDR picture is greater than or equal to `MaxPicOrderCntLsb/2`, it may be constrained such that signaling of POC MSB information for the IRAP pictures or the GDR picture is present. Here, when the value of `ph_recovery_poc_cnt` associated with the GDR picture is 0, the GDR picture itself may be recovery point picture.

[0394] The information on the POC encoded in step S1230 may be transmitted to an image decoding apparatus through a bitstream. In this case, the bitstream may be stored in a non-transitory computer-readable recording medium.

[0395] FIG. 13 is a flowchart illustrating an image decoding method according to an embodiment of the present disclosure.

[0396] The image decoding apparatus of FIG. 13 may be performed by the image decoding apparatus of FIG. 3 or FIG. 6. For example, step S1310 to step S1330 may be performed by the entropy decoder 210, 610-0 or 610-1 and the prediction unit 260, 630-0 or 630-1.

[0397] Referring to FIG. 13, the image decoding apparatus may determine whether first information specifying a value of a POC (picture order count) MSB (Most Significant Bit) cycle of a current picture is present in a picture header (S1310). In the present disclosure, the first information may be referred to as POC MSB information.

[0398] In an example, the image decoding apparatus may determine whether the first information (e.g., `ph_poc_msb_cycle_val`) is present in the picture header, based on `sps_poc_msb_cycle_flag` obtained from an SPS (sequence parameter set). For example, when `sps_poc_msb_cycle_flag` has a first value (e.g., 1), the image decoding apparatus may determine whether the first information is present in the picture header, based on `ph_poc_msb_cycle_present_flag` obtained from the picture header (e.g., presence when `ph_poc_msb_cycle_present_flag` is 1/absence when `ph_poc_msb_cycle_present_flag` is 0). In contrast, when `sps_poc_msb_cycle_flag` has a second value (e.g., 0), the image decoding apparatus may determine that the first information is not present in the picture header.

[0399] In an embodiment, based on the current picture being an IRAP (intra random access point) picture or a GDR picture (i.e., a GDR picture with associated `ph_recovery_poc_cnt` equal to 0) as a recovery point picture, the first information may be present in the picture header. In this case, the IRAP picture or the GDR picture may not be a CLVSS (coded layer video sequence start) picture in an independent non-output layer. In addition, only IRAP pictures and a GDR picture in a current layer including the current picture may be used for inter layer prediction by layers different from the current layer. In addition, a distance between the current picture and a previous IRAP picture preceding the current picture in decoding order or a GDR picture as a recovery point picture may be greater than or equal to half a maximum value (e.g., `MaxPicOrderCntLsb`) of the POC LSB.

[0400] The above-described embodiment may be defined as a constraint for bitstream conformance as in Embodiment 9 of the present disclosure. That is, as IRAP pictures included in a layer which is not an output layer as an independent layer or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0, only the IRAP pictures and the GDR picture are required for inter layer prediction by other layer(s) and, when a distance between

the IRAP pictures or the GDR picture and a previous IRAP picture or GDR picture is greater than or equal to $\text{MaxPicOrderCntLsb}/2$, it may be constrained such that signaling of POC MSB information for the IRAP pictures or the GDR picture is present. Here, when the value of `ph_recovery_poc_cnt` associated with the GDR picture is 0, the GDR picture itself may be recovery point picture.

[0401] The image decoding apparatus may determine the POC MSB of the current picture, based on whether the first information is present in the picture header (S1320). In an example, when the first information (e.g., `ph_poc_msb_cycle_val`) is present in the picture header, the POC MSB may be derived by multiplying the value of the POC MSB cycle by a maximum value (e.g., `MaxPicOrderCntLsb`) of a POC LSB (Least Significant Bit) of the current picture. In contrast, when the first information is not present and the current picture is a CLVSS (coded layer video sequence start) picture, the POC MSB may be derived to be 0. In the other case, the POC MSB may be derived based on a POC MSB (e.g., `prevPicOrderCntMsb`) of a previous picture preceding the current picture in decoding order, as described above with reference to Table 2.

[0402] The image decoding apparatus may derive the POC of the current picture based on the POC MSB determined in step S1320 (S1330). In an example, the POC (e.g., `PicOrderCntMsb`) may be derived by adding the POC LSB (e.g., `ph_pic_order_cnt_lsb`) of the current picture to the POC MSB (e.g., `PicOrderCntMsb`).

[0403] In an example, based on the first information being not present in the picture header and the current picture being not a CLVSS picture, a POC difference between the current picture and a previous picture (e.g., `Tid0Pic`) preceding the current picture in decoding order may be less than half a maximum value (e.g., `MaxPicOrderCntLsb`) of a POC LSB of the current picture. In this case, a temporal identifier (e.g., `TemporalId`) of the previous picture may be 0. In addition, a layer identifier (e.g., `nuh_layer_id`) for a NAL unit of the previous picture may be equal to a layer identifier for a NAL unit of the current picture, and the previous picture may be used as a reference picture (e.g., `ph_non_ref_pic_flag=0`) and may not be a RASL (Random Access Skipped Leading) picture or a RADL (Random Access Decodable Leading) picture. The above-described embodiment may be defined as in Implementation 7.

[0404] As described above, according to the image encoding/decoding method according to an embodiment of the present disclosure, which is described above with reference to FIGS. 13 and 14, for an IRAP picture and/or a GDR picture with an associated `ph_recovery_poc_cnt` value equal to 0, signaling of POC MSB information (e.g., `ph_poc_msb_cycle_val`) may be forced under a predetermined condition. In this case, a distance between the IRAP picture and/or the GDR picture and a previous IRAP picture or GDR picture may be less than or equal to $\text{MaxPicOrderCntLsb}/2$. Therefore, when a non-output layer includes only the IRAP picture and/or the GDR picture (see FIG. 9), as a decoder cannot find a previous picture (i.e., `prevTid0Pic`) of the IRAP picture and/or the GDR picture in decoding order, the problem that the POC of the IRAP picture and/or the GDR picture is not derived correctly may be solved.

[0405] While the exemplary methods of the present disclosure described above are represented as a series of operations for clarity of description, it is not intended to limit the order in which the steps are performed, and the

steps may be performed simultaneously or in different order as necessary. In order to implement the method according to the present disclosure, the described steps may further include other steps, may include remaining steps except for some of the steps, or may include other additional steps except for some steps.

[0406] In the present disclosure, the image encoding apparatus or the image decoding apparatus that performs a predetermined operation (step) may perform an operation (step) of confirming an execution condition or situation of the corresponding operation (step). For example, if it is described that predetermined operation is performed when a predetermined condition is satisfied, the image encoding apparatus or the image decoding apparatus may perform the predetermined operation after determining whether the predetermined condition is satisfied.

[0407] The various embodiments of the present disclosure are not a list of all possible combinations and are intended to describe representative aspects of the present disclosure, and the matters described in the various embodiments may be applied independently or in combination of two or more.

[0408] Various embodiments of the present disclosure may be implemented in hardware, firmware, software, or a combination thereof. In the case of implementing the present disclosure by hardware, the present disclosure can be implemented with application specific integrated circuits (ASICs), Digital signal processors (DSPs), digital signal processing devices (DSPDs), programmable logic devices (PLDs), field programmable gate arrays (FPGAs), general processors, controllers, microcontrollers, microprocessors, etc.

[0409] In addition, the image decoding apparatus and the image encoding apparatus, to which the embodiments of the present disclosure are applied, may be included in a multimedia broadcasting transmission and reception device, a mobile communication terminal, a home cinema video device, a digital cinema video device, a surveillance camera, a video chat device, a real time communication device such as video communication, a mobile streaming device, a storage medium, a camcorder, a video on demand (VOD) service providing device, an OTT video (over the top video) device, an Internet streaming service providing device, a three-dimensional (3D) video device, a video telephony video device, a medical video device, and the like, and may be used to process video signals or data signals. For example, the OTT video devices may include a game console, a blu-ray player, an Internet access TV, a home theater system, a smartphone, a tablet PC, a digital video recorder (DVR), or the like.

[0410] FIG. 14 is a view showing a content streaming system, to which an embodiment of the present disclosure is applicable.

[0411] As shown in FIG. 14, the content streaming system, to which the embodiment of the present disclosure is applied, may largely include an encoding server, a streaming server, a web server, a media storage, a user device, and a multimedia input device.

[0412] The encoding server compresses content input from multimedia input devices such as a smartphone, a camera, a camcorder, etc. into digital data to generate a bitstream and transmits the bitstream to the streaming server. As another example, when the multimedia input devices such as smartphones, cameras, camcorders, etc. directly generate a bitstream, the encoding server may be omitted.

[0413] The bitstream may be generated by an image encoding method or an image encoding apparatus, to which the embodiment of the present disclosure is applied, and the streaming server may temporarily store the bitstream in the process of transmitting or receiving the bitstream.

[0414] The streaming server transmits the multimedia data to the user device based on a user's request through the web server, and the web server serves as a medium for informing the user of a service. When the user requests a desired service from the web server, the web server may deliver it to a streaming server, and the streaming server may transmit multimedia data to the user. In this case, the content streaming system may include a separate control server. In this case, the control server serves to control a command/response between devices in the content streaming system.

[0415] The streaming server may receive content from a media storage and/or an encoding server. For example, when the content is received from the encoding server, the content may be received in real time. In this case, in order to provide a smooth streaming service, the streaming server may store the bitstream for a predetermined time.

[0416] Examples of the user device may include a mobile phone, a smartphone, a laptop computer, a digital broadcasting terminal, a personal digital assistant (PDA), a portable multimedia player (PMP), navigation, a slate PC, tablet PCs, ultrabooks, wearable devices (e.g., smartwatches, smart glasses, head mounted displays), digital TVs, desktops computer, digital signage, and the like.

[0417] Each server in the content streaming system may be operated as a distributed server, in which case data received from each server may be distributed.

[0418] The scope of the disclosure includes software or machine-executable commands (e.g., an operating system, an application, firmware, a program, etc.) for enabling operations according to the methods of various embodiments to be executed on an apparatus or a computer, a non-transitory computer-readable medium having such software or commands stored thereon and executable on the apparatus or the computer.

[0419] The embodiments of the present disclosure may be used to encode or decode an image.

What is claimed is:

1. A decoding apparatus for image decoding, the decoding apparatus comprising:

- a memory; and
- at least one processor connected to the memory, the at least one processor configured to:
 - obtain a first flag specifying whether a second flag is present in a picture header;
 - obtain the second flag specifying whether first information is present in the picture header based on the first flag specifying that the second flag is present in the picture header;
 - determine whether the first information specifying a value of a picture order count (POC) most significant bit (MSB) cycle of a current picture is present in the picture header based on the second flag;
 - determine a POC MSB of the current picture based on whether the first information is present in the picture header; and
 - derive a POC of the current picture based on the determined POC MSB,
- wherein, based on the first information being not present in the picture header and the current picture being not

a CLVSS (coded layer video sequence start) picture, a POC difference between the current picture and a previous picture preceding the current picture in decoding order is less than half a maximum value of a POC least significant bit (LSB) of the current picture, and wherein, based on the current picture being an IRAP (intra random access point) picture or a GDR (gradual decoding refresh) as a recovery point picture, the first information is present in the picture header.

2. The decoding apparatus of claim 1, wherein a temporal identifier of the previous picture is 0.

3. The decoding apparatus of claim 1, wherein a layer identifier for a NAL unit of the previous picture is equal to a layer identifier of a NAL unit of the current picture, and

wherein the previous picture is able to be used as a reference picture and is not a RASL (random access skipped leading) picture or a RADL (random access decodable leading) picture.

4. The decoding apparatus of claim 1, wherein the IRAP picture or the GDR picture is not a CLVSS picture in an independent non-output layer.

5. The decoding apparatus of claim 1, wherein only IRAP pictures and a GDR picture in a current layer including the current picture are used for inter layer prediction by layers different from the current layer.

6. The decoding apparatus of claim 1, wherein a distance between the current picture and a previous IRAP picture preceding the current picture in decoding order or a GDR picture as a recovery point picture is greater than or equal to half a maximum value of the POC LSB.

7. An encoding apparatus for image encoding, the encoding apparatus comprising:

- a memory; and
 - at least one processor connected to the memory, the at least one processor configured to:
 - derive a picture order count (POC) of a current picture;
 - determine POC most significant bit (MSB) of the current picture based on the derived POC; and
 - encode information on the derived POC,
 - wherein a first flag specifying whether a second flag is present in a picture header is encoded,
 - wherein the second flag specifying whether first information is present in the picture header is encoded based on determining that the second flag is present in the picture header,
 - wherein the first information specifying a value of a POC MSB cycle of the current picture is encoded based on determining that the first information is present in the picture header,
 - wherein, based on the first information being not encoded and the current picture being not a CLVSS (coded layer video sequence start) picture, a POC difference between the current picture and a previous picture preceding the current picture in decoding order is less than half a maximum value of a POC least significant bit (LSB) of the current picture, and
 - wherein, based on the current picture being an IRAP (intra random access point) picture or a GDR (gradual decoding refresh) as a recovery point picture, the first information is encoded in a picture header.
8. The encoding apparatus of claim 7, wherein a temporal identifier of the previous picture is 0.

9. The encoding apparatus of claim 7, wherein a layer identifier for a NAL unit of the previous picture is equal to a layer identifier of a NAL unit of the current picture, and wherein the previous picture is able to be used as a reference picture and is not a RASL (random access skipped leading) picture or a RADL (random access decodable leading) picture.
10. The encoding apparatus of claim 7, wherein the IRAP picture or the GDR picture is not a CLVSS picture in an independent non-output layer.
11. The encoding apparatus of claim 7, wherein only IRAP pictures and a GDR picture in a current layer including the current picture are used for inter layer prediction by layers different from the current layer.
12. An apparatus for transmitting data for an image, the apparatus comprising:
at least one processor configured to obtain a bitstream generated by an encoding method performed by an encoding apparatus for image encoding; and
a transmitter configured to transmit the bitstream, wherein the encoding method comprising:
deriving a picture order count (POC) of a current picture;
determining POC most significant bit (MSB) of the current picture based on the derived POC; and

- encoding information on the derived POC,
wherein a first flag specifying whether a second flag is present in a picture header is encoded,
wherein the second flag specifying whether first information is present in the picture header is encoded based on determining that the second flag is present in the picture header,
wherein the first information specifying a value of a POC MSB cycle of the current picture is encoded based on determining that the first information is present in the picture header,
wherein, based on the first information being not encoded and the current picture being not a CLVSS (coded layer video sequence start) picture, a POC difference between the current picture and a previous picture preceding the current picture in decoding order is less than half a maximum value of a POC least significant bit (LSB) of the current picture, and
wherein, based on the current picture being an IRAP (intra random access point) picture or a GDR (gradual decoding refresh) as a recovery point picture, the first information is encoded in a picture header.

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